TM Forum Specification

Open Gateway Operate API - Product Catalog User Guide

TMF936

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Introduction

The following document is intended to guide developers and implementation on catalog definition and synchronization in the context of the GSMA Open Gateway initiative.

This Operate API covers the Product Catalog management as described by GSMA for Open Gateway context. This API focusses on :

- Product Offering API Management (retrieving)
- Product Specification API Management (retrieving)

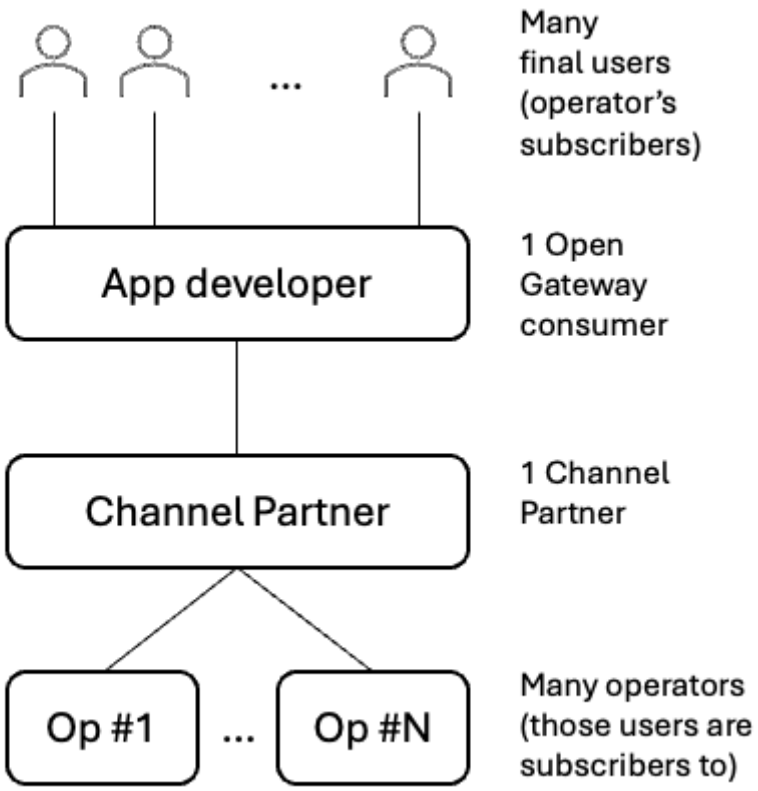
This API is a DCS (Domain Context Specialization) of TMF620 Product Catalog API v5.0 for Open Gateway context.

Disclaimer: All prices examples appearing in this specification are fictitious. Any resemblance to real offers is purely coincidental.

Channel Partner model overview

Business requirements for the definition of this API come from GSMA Open Gateway initiative, in particular this document : https://www.gsma.com/solutions-and-impact/gsma-open-gateway/wp-content/uploads/2024/02/Channel-Partner-Onboarding-Guide-WA.101-v1.0.pdf.

Open Gateway Channel Partner model is composed of different actors:

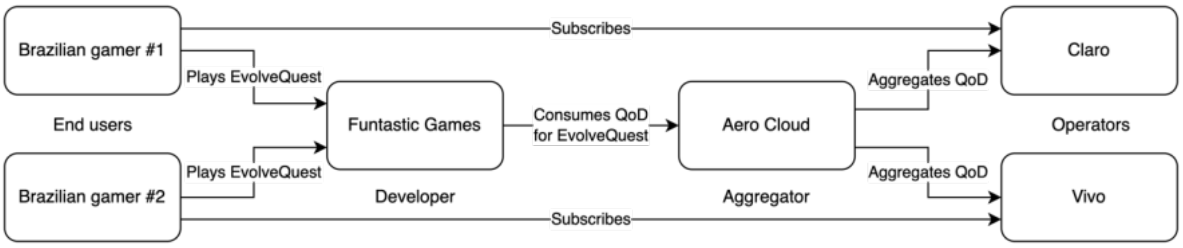


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- End Users who are both subscribers to Operators and users of the application published by the Developer
- App Developer (ApplicationOwner) create applications that are acquired and utilized by end users
- Channel Partners aggregate/expose Operator's CAMARA standardised APIs to build Open Services and implement Operator end-point routing based on final user identification on the networks.
- Operators expose telco capabilities through CAMARA standardised APIs and could partner with Channel Partners to enable the Services that they offer to Developers.

The following is an example of interaction between the actors on the model:

"Funtastic Games" is a company developing and publishing videogames (the Developer) and hosts its servers on the cloud services from the cloud provider "Aero Cloud" which commercialises Services as well (the Channel Partner). Funtastic Games wants to publish a game (the application) called "EvolveQuest" which features the network capability to assign custom quality when used in cloud gaming mode on mobile devices (Open Gateway QualityOnDemand product). Two different gamers playing EvolveQuest in Brazil are subscribers to different Operator's mobile network access services, resulting in the following relational diagram:



Operate APIs

As indicated, the Channel Partner indirect model enables a single channel for tenant applications to gain access to capabilities from multiple Operators, without the need for the customer to set up a contractual relationship with each Operator. It implies the need to provide certain additional functionalities from the Operators to the Channel Partner to enable the commercial model.

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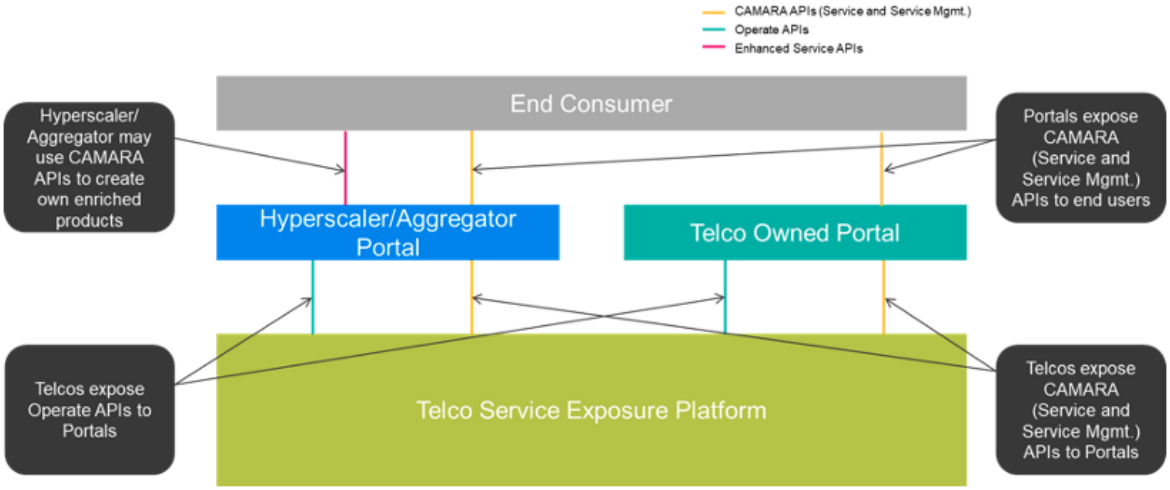


Figure 3. Families of APIs involved in Open Gateway

Therefore, in addition to the CAMARA APIs, the Operate APIs are proposed for offering programmable access to Operation, Administration and Management (OAM) capabilities to facilitate the integration of the Open Gateway NaaS Platform with Channel Partner. These APIs provide all the transversal/non-service specific functionalities that are required to make a commercial offering out of the Service APIs and Service Management APIs, making them operable and monetizable. Examples of functions provided by the Operate APIs include ordering, monitoring, trouble ticketing, accounting, billing, payment, etc. and also API discovery, user registration and consent management, etc.

Scope

The proposal for this Operate API is focused on defining the specific functionality in the Operate APIs that enables the management of the catalog of Offerings (APIs) between Channel Partners and Operators. This functionality is focused on enabling automation when managing a common interface to the catalog between the different Operators (producers of the catalog offerings) and the channel partner (consumer of the offerings), so a synchronization is always ensured when new offerings are created, or when an existing offering is updated or deprecated.

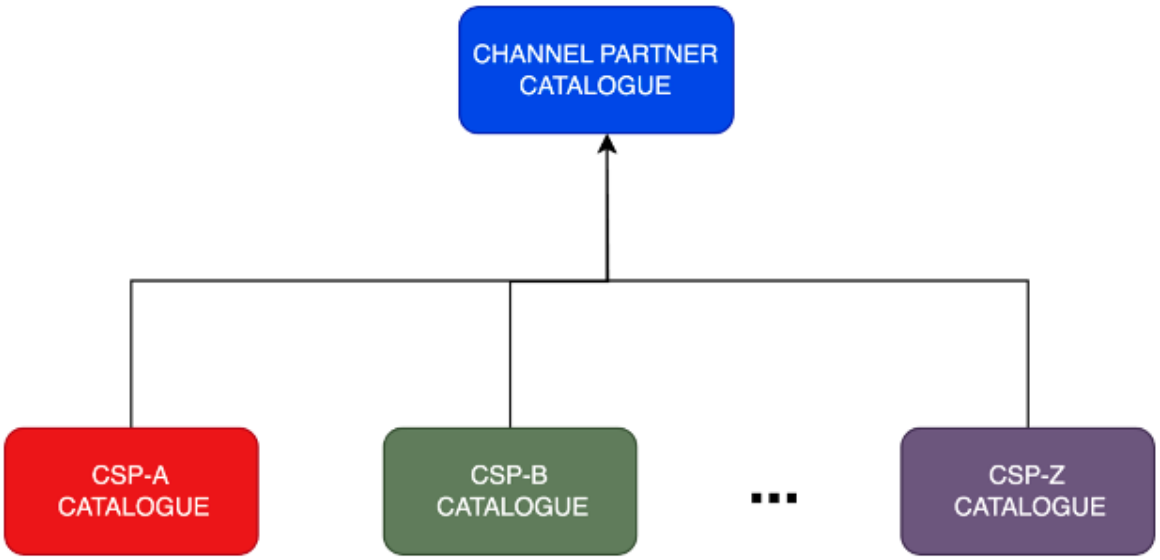
Operators provide Channel Partners with offerings, each of them is defined as the combination of:

- A CAMARA standardized API available on the Operator's API exposure platform.
- A concrete CAMARA standardized API scope endpoint/functionality, defined as a specific API technical endpoint scope of a CAMARA API. For example. "location-verification", "
- A usage purpose (or a list of possible usage purposes, the purpose value being configurable during the ordering), based on a set of standardized W3C purposes (https://w3c.github.io/dpv/dpv/#vocab-purpose) that the offering can be used for.
- Charging models and a pricing list (oneShot, recurring, usage based, …).
- Additional optional bundled offering (usage volume, customer support…)

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- Different characteristics associated to each APIs, that can be used to configure offerings during the ordering.
- A description of the offering and its value proposition.
- The minimum and maximum versions of the CAMARA API supported by the Operator.
- Specific terms & conditions of use, including SLAs and privacy requirements associated to the purpose.

Additionally, Channel Partners, as Open Gateway offerings Customer/Distributor, will also host and manage their own catalog beyond the offerings from the different Operators that are exposed by the Channel Partner.



The purpose of this TM Forum API specification is to propose a data model to define API offerings (includes TMF entities ProductOffering and ProductSpecification) that will enable synchronization between the catalog of a Operator and a Channel Partner's catalog, and the catalog of the Channel partner

The Channel partner will require two different set of offerings as part of its internal catalog:

1. One set of offerings that include the offerings from each Operator.
2. One set of offerings that include the proper offering that the Channel Partner is exposing to its own customers. Depending on the aggregation model, Channel Partner will create this catalog with:
    a. Marketplace Aggregator: Directly a replica of the Operator's offerings, aligning the characteristics for being able to provide an aggregated offering. e.g. Location Verification OGW API.
    b. Wholesale Reseller: A Channel Partner own offering, which is composed by (at least) the aligned OGW offering of all the integrated Operators. e.g. Geolocation API, composed by OGW Location Verification and others.

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ODA Component

At the time this document version has been released, no ODA component exposes this API.

Information Framework

This API specification is a specialization of TMF620 Product Catalog API, so relies on Product Offering Specification, Product Specification and Product Offering Price SID ABE (Aggregate Business Entities). No other SID ABE have been used as a basis for this API definition.

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Sample Use Cases

This Operate API takes into account both models defined in GSMA Playbook :

In the Wholesale Reseller model, the service is fully managed by the Channel Partner. In the Aggregated Marketplace model, the Channel Partner leverages its Marketplace to individually expose the APIs. To ensure a coherent user experience, a UX layer is applied, creating a unified interface for the Developer.

For both cases, the Catalog API supports two mechanisms for synchronization:

- Push mode: Channel Partner subscribes to Operator's catalog events. Notifications are sent by Operator and received by Channel Partner in case any modification is implemented, so Channel partner is informed about any change in the Operator's offering.
- Pull mode: Channel Parner proactively queries for offers, either for all offers, or possibly with filters on: creation date (e.g. all new offer since last month), name (e.g. all offers on Quality on Demand (QoD)) or other fields.

ODA End to end Use Cases

The document TMFS026 (under IG1228 work) describes how to use this API within ODA (Open Digital Architecture) context, in particular ODA Components, using a realistic example. This document is updated regularly.

Limitations of this specification version

This first version of this Operate API exposes only GET Operations on atomic and simple bundles of ProductOffering (using BundledProductOffering schema instead of BundledGroupProductOffering schema) and ProductSpecification. ProductOfferingPrices describe prices with a non machine readable format; because there is not yet business requirements for this. So the 3 types of ProductOfferingPrice possible : usage, oneTime and recurring are composed within a single productOfferingPrice where the definition of the price is given in a textual description. Prices alterations are also not implemented (discount, tax…) and can be included in the description of the price. So prices values are not linked in the json payload to product characteristic values (example on Usage Volume offering : if volume = 1000 API calls then price value is x EUR, if volume = 500 API calls, price value is y EUR).

This version only propose actions add, modify and delete. Future versions may include migrate, for migration of ProductOffering or of ProductSpecification, if requirements confirm this need.

ProductSpecification and ProductOfferingUsagePriceCharge don't contain yet the reference of the productUsageSpecification because this resource will be available in the DCS Product Usage API, not yet specified.

Categories are not needed for now.

Agreement between Channel Partner and Operator for Open Gateway is not yet represented in the API. It would be useful to permit a Channel Partner that have 2 agreements with same Operator to filter the catalog based on the Agreement id.

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For now the need for a qualification to help Channel Partner to retrieve the offer it is eligible to is not confirmed. This specification is simple, giving an access for Channel Partner to the Operator catalog. Channel Partner is supposed to be identified in the API call so that the Operator answer is only a restrictive view of what the Channel Partner is allowed to view.

A need to be confirmed for qualification would be this one : a Customer wants to know which ProductOffering is eligible for order based on a context, for instance, for a specific Application, the result could be the whole Catalog of offers except the ProductOffering that were already order by the Customer for this Application, or any restriction on the response guided by another rule (like if Customer owns this product, it can't order this one again). For now, in the absence of such a qualification process, the product order will respond with an error is the order is made for a non eligible offer.

This version of the specification proposes only notifications for ProductOffering. As ProductOffering contains reference to all other subresources passed by reference (at least ProductSpecification) with their version, each change on these referred resources that is changing the value of one of its attributes, imply the change of the version of this resource. So this change of version of the referred resource (for instance the ProductSpecification) is visible into the ProductOffering payload, so may generate a notification ProductOfferingAttributeValueChangeEvent. ProductOfferingVersionCreateEvent is not yet proposed in this API.

Requirement for cascading of Channel Partner (a Channel Partner exposing TMF936 to another Channel Partner) is not taken into account yet.

There are at least 2 kinds of ProductSpecification that are identified :

- ApiProductSpecification : this is based on a Customer Facing Service providing a service accessible by API. Typically a single CAMARA API.
- UsageVolumeProductSpecification : this allows to manage balance (or bucket) of usage that apply to one or several ApiProductSpecification on a set of specific ProductUsageSpecification. The relationship between the ProductSpecification and the ProductUsageSpecification of the concerned ProductSpecification is not yet described, waiting for the Product Usage API to be released.

Documentation of ProductSpecification is using a specialization of the standard Attachment schema, for best interoperability.

Further versions of this API specification may take into account these above limitations if the requirements are confirmed.

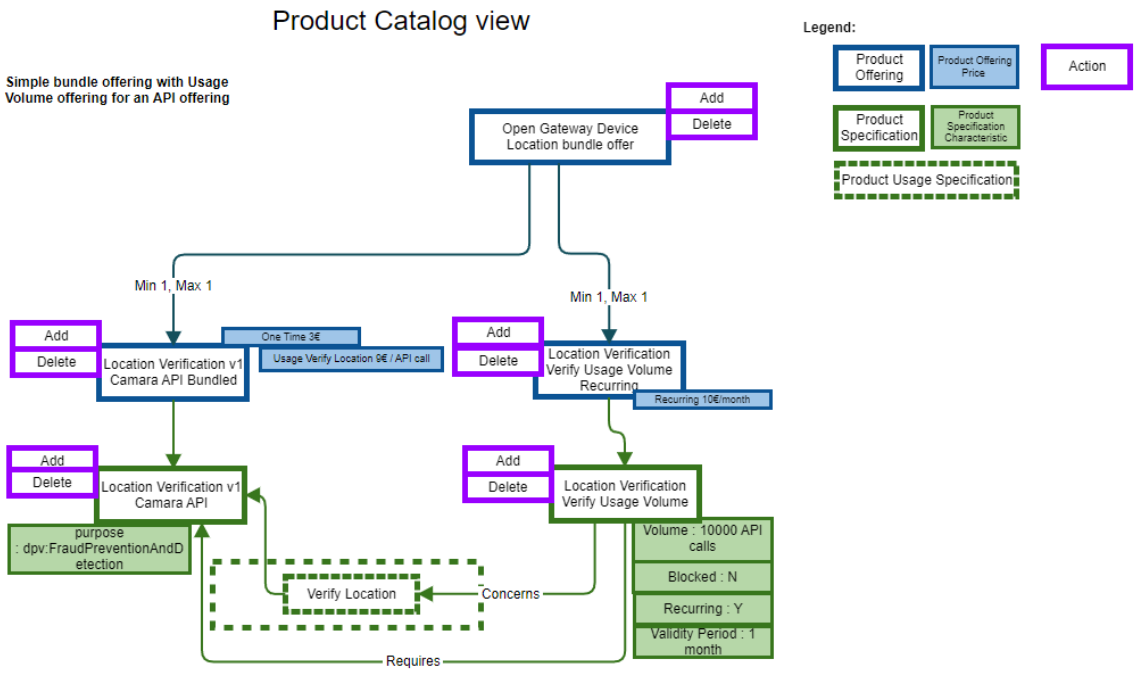
Catalog example

The examples in the OAS and User Guide are based on this ficticious catalog model.

Disclaimer: All prices examples appearing in this specification are fictitious. Any resemblance to real offers is purely coincidental.

Bundle offer (Operator internal view) :

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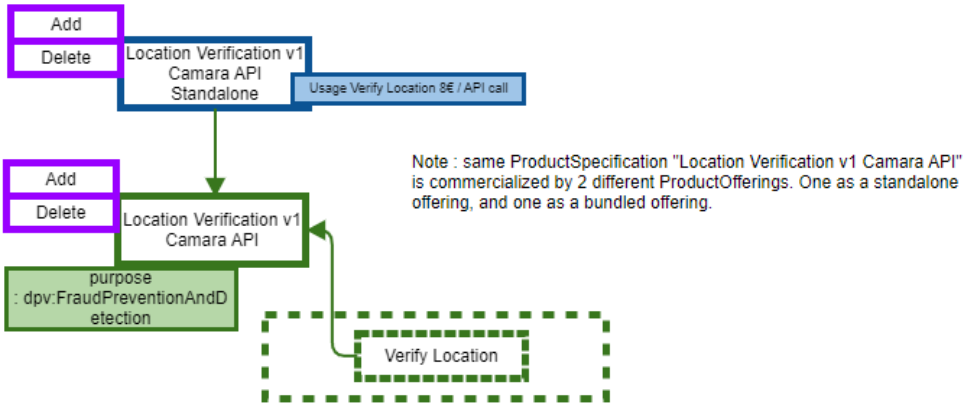


The product catalog offers for Open Gateway Location Verification a flexible model based on subscription and usage volume for an API verification service. The main offering is the Open Gateway bundle offer, which allows users to add or remove components based on their specific needs. This structure provides an adaptable solution for different use cases.

Within this offering, there are two main location verification included offerings. The first one, Location Verification v1 CAMARA API Bundled, includes the Camera API as part of a package and is specifically marketed for fraud prevention and detection.

The second included offering, Location Verification Verify Usage Volume Recurring, follows a usage-based model with recurring cost, enabling continuous access to the service. Both included offerings can be configured dynamically based on user requirements.

Standalone offer (Operator internal view) :



The concepts of ProductSpecification and ProductOffering permits Channel Partner to know which are the offerings made upon a same unique ProductSpecification.

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An Operator can design as many atomic ProductOffering as he needs based on a same ProductSpecification. An atomic ProductOffering can be based on maxiumum 1 ProductSpecification. A composite ProductOffering (bundle, package, …) is composed of atomic ProductOfferings or composite ProductOfferings.

Important notice : it has been decided to simplifiy this view for the Channel Partner. We consider that the action (add, delete, modify) done at the atomic productOffering level applies same at its related productSpecification level. This implies that the product instantiated from an atomic productOffering order (action add) aggregates the instance of the atomic productOffering and the instance of the productSpecification. Inside the Operator IT it could be 2 separated products, but the view given to Channel Partner is only one product. For a composite productOffering, each productOffering that is part of the composition (including the main composite offer itself) has an instantiated product created for each productOffering that has an action add on the order. It also implies that the product order contains only the id of the productOfferings (not the id of the productSpecifications) but their productAction contain productSpecification characteristics, so apply to the ProductSpecification.

In the examples below, the id are the last 3 characters of the full id present in the OAS examples of TMF936 and TMF931. This aims to provide comprehensive understanding of the OAS examples based on these diagrams. Please note that the ProductOrder examples in OAS are not represented in these diagrams. In TMF931 there are :

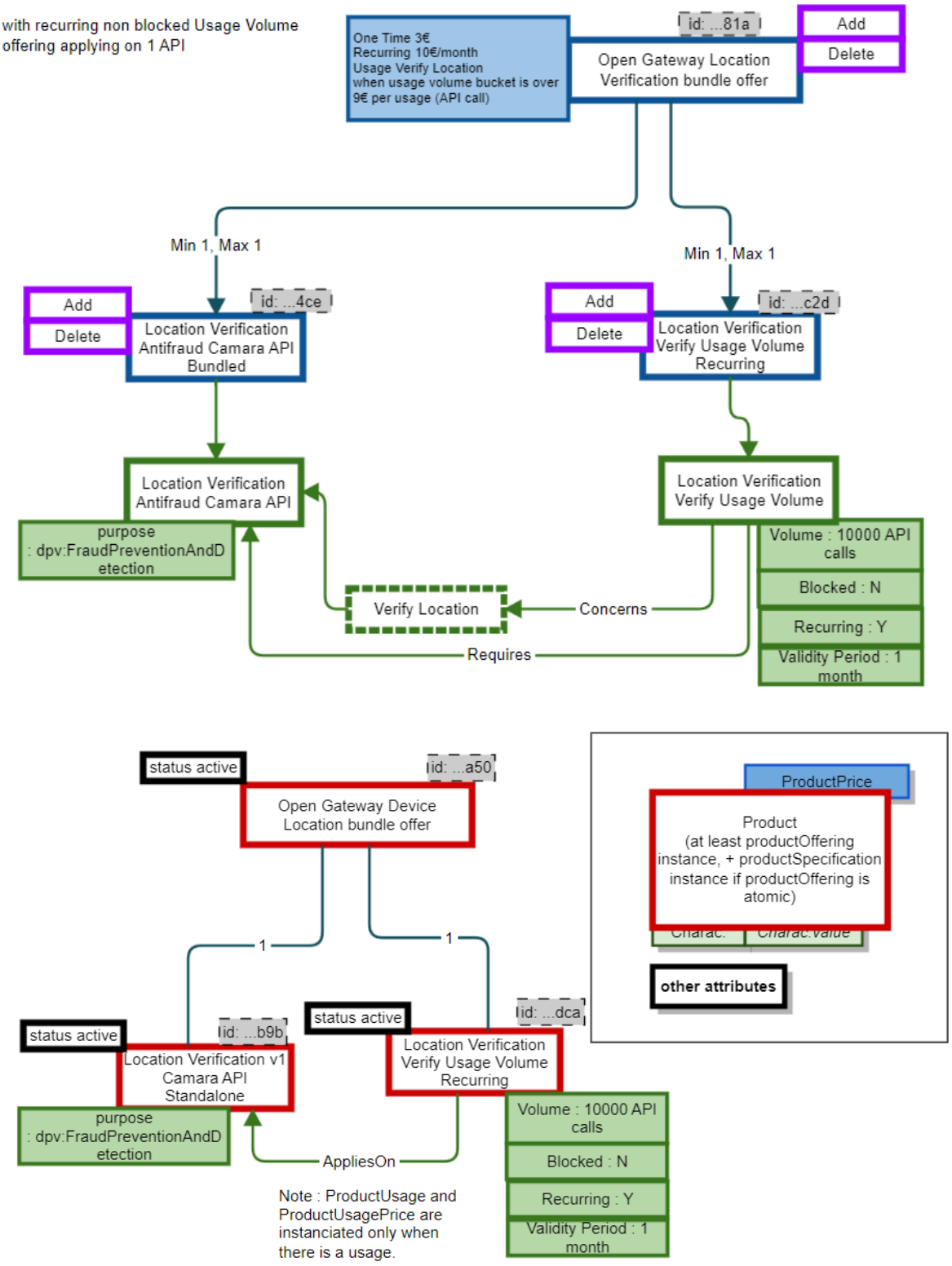
- 1 example of order with action add for the standalone ProductOffering/ProductSpecification
- 1 example of order with action delete for the standalone Product (instance of the standalone ProductOffering/ProductSpecification)
- 1 example of order with 3 order items : 1 item with action add for the bundle ProductOffering/ProductSpecification, 1 item with action add for the atomic API ProductOffering/ProductSpecification and 1 item with action add for the atomic usage volume ProductOffering/ProductSpecification. These 3 order items are in relationship, keeping same relationships as in the diagram. The 3 order items are mandatory to fulfil the order. This permits to pass the necessary characteristics (in ProductAction) and other attributes that are specific for each ProductOffering/ProductSpecification.

Bundle offer (Channel Partner view of catalog and inventory) :

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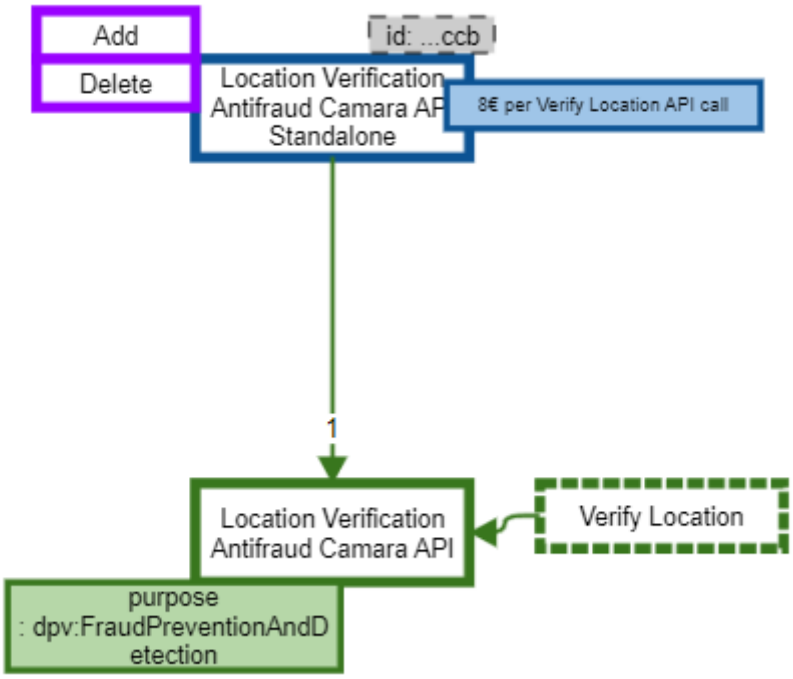
Bundle offering

with recurring non blocked Usage Volume offering applying on 1 API

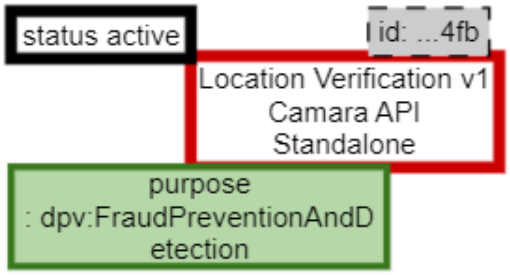


Standalone offer (Channel Partner view of catalog and inventory) :

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Standalone API offering



Note : as per choice to not expose Product Order Action Item at ProductSpecification level, the instanciation of Atomic ProductOffering and Atomic ProductSpecification are aggregated into 1 single Product

Language

Multi language is supported using the optional standard Accept-Language header. See TMF630 Design Guidelines. Each API call will return payload for only 1 language as provided by the client. Default language is English.

Push mode

Operator provides a notification (hub) resource to permit API client (Channel Partner) to subscribe and receive events sent by the Catalog component. The following events are covered in the API

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- ProductOffering create event When an Operator creates a new ProductOffering.
- ProductOffering attribute (property) value change event: When an existing ProductOffering is updated in the Operator.
- ProductOffering deletion event: When an existing ProductOffering is deleted in the Operator.
- ProductOffering state change event: When an existing ProductOffering lifecycleStatus is updated in the Operator.

Note : The choice has been made that the Channel Partner is interested first in the Product Offering changes. A change on the ProductSpecification should change the ProductSpecification version. This ProductSpecification version change should force the change of the ProductOfferings versions that are built above this ProductSpecification.

Pull mode

These steps are to be considered when a Channel Partner wants to retrieve the list of offer(s) of an Operator, or wants the details of an specific offer.

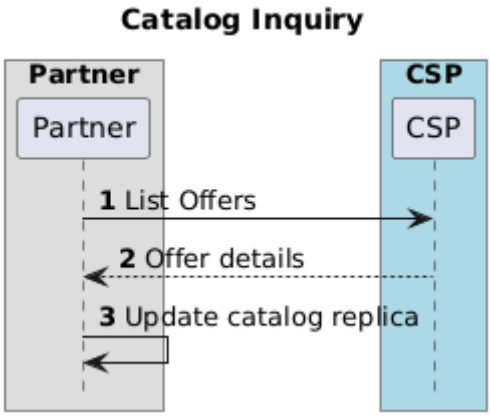
- Retrieve complete offer list.
- Retrieve specific offer details.

Use-Cases

Utilizing the above mentioned Capabilities the following use-cases can be realized

UC 1: Full offerings Syncronization

On Initial onboarding (or upon need) Channel partners will inquire the CSPs for full offering details. This use-case expects both consumers and providers to be able to handle full payload of all available offerings



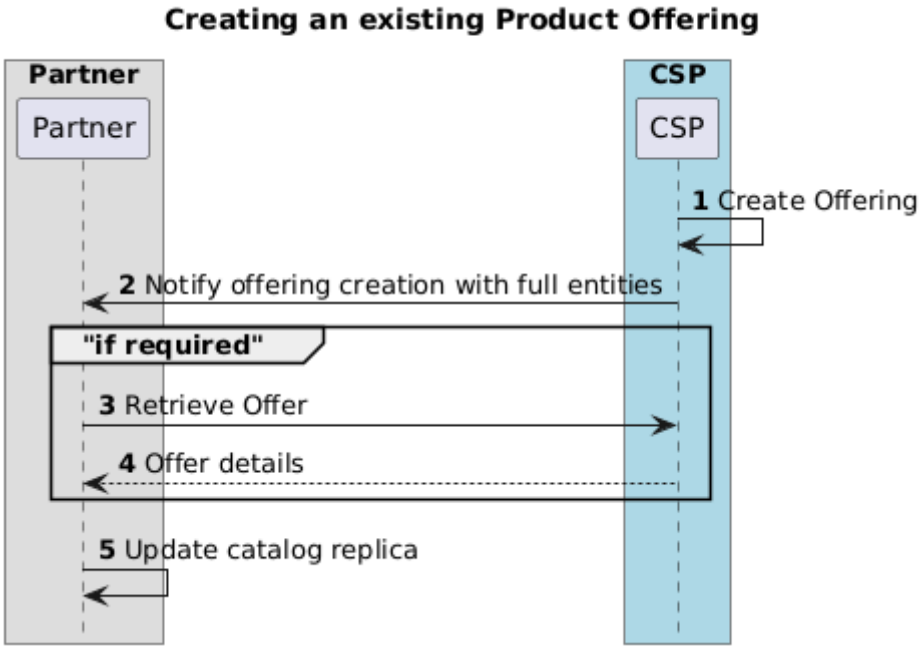
Operation :

- GET /productOffering

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UC 2: New Product Offering Synchronization

On Creation of a new product offering CSPs would notify the subscribed Channel Partners of the creation of new offering. The Channel Partner can then inquire the CSP for the details of the created offering using the ID recieved upon notification



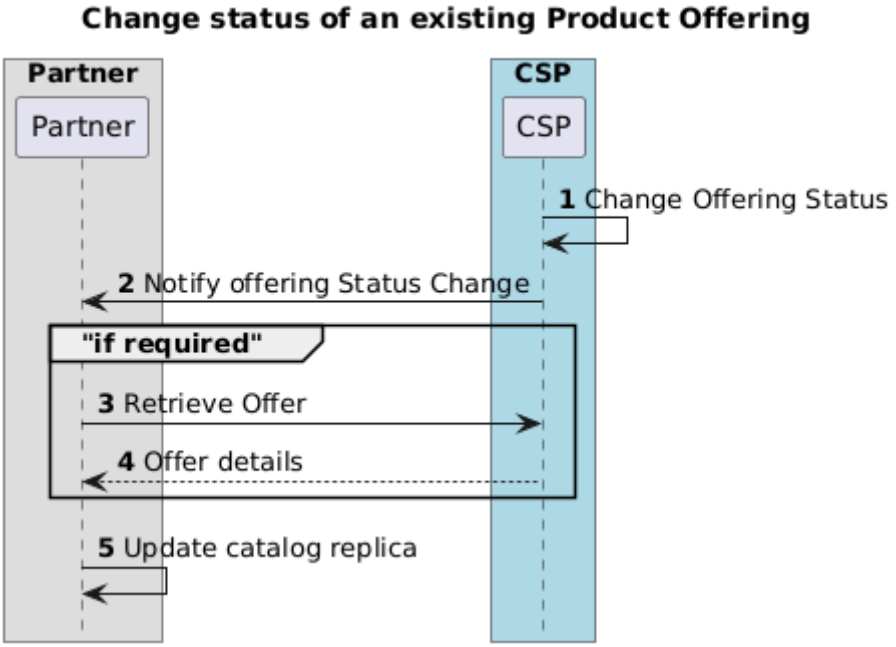
Operation :

- ProductOfferingCreateEvent
- GET /productOffering/{id}

UC 3: Deactivating or Suspending an existing Product Offering

CSP modifies product offering status would notify the subscribed Channel Partners of the status change action. The channel partners can optionally inquire for more details of the offering using the ID recieved upon notification

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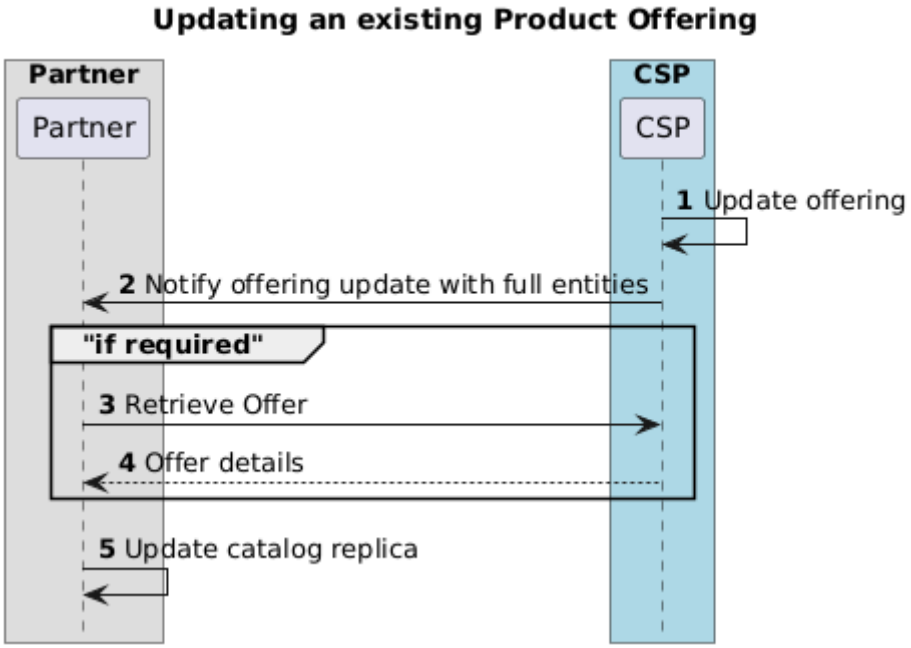


Operation :

- ProductOfferingStateChangeEvent
- GET /productOffering/{id}

UC 4: Updating an existing Product Offering

CSP modifies product offering attributes or associated entities (Product Offering Price, TermsOrConditions, Product Specification) updates



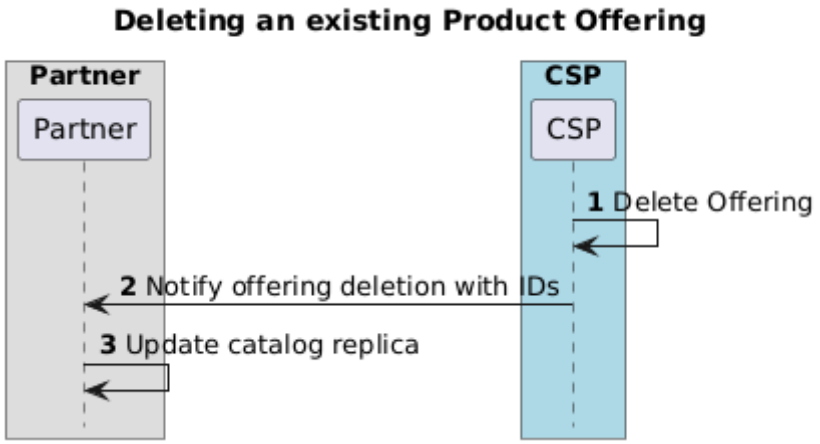
Operation :

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- ProductOfferingAttributeValueChangeEvent
- GET /productOffering/{id}

UC 5: Deleting an existing Product Offering

CSP deletes product offering (removing it from the catalog) would notify the subscribed Channel Partners of the status change action.

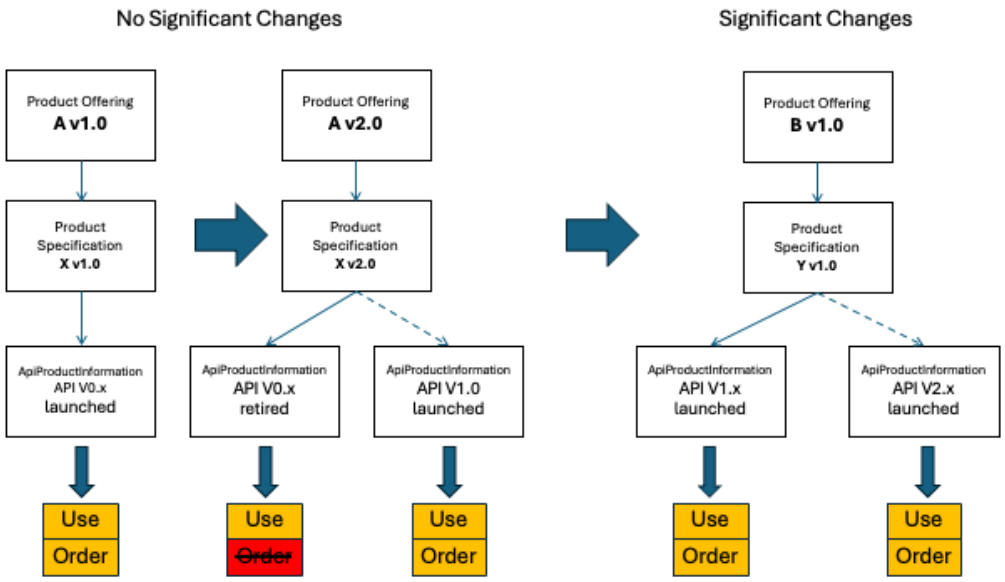


Operation :

- ProductOfferingDeleteEvent

API Versioning

- Support to different API versions at ApiProductSpecification



ApiProductSpecification model SHALL support the typical lifecycle of API specifications:

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- ApiProductSpecifications MAY evolve organically by adding or removing supported API versions in time. Organical evolution may be applied if new API versions do not add significant privacy requirements or feature changes, otherwise new ApiProductSpecification and OpenGatewayProductOffering creation should be considered by product managers.
- Each API version may have specific characteristics (path, scopes, purpose, authorization grant types, etc.).
- The catalog model SHALL support at API version level the concept of API lifecycle: An API version MAY be declared active (it can be provisioned in TMF931 Onboarding API) or retired (it can be used by Applications already provisioned but cannot be provisioned for new Applications).
- OpenGatewayProductOffering, ApiProductSpecification and API version definitions SHALL be able to use the resouce versioning model (See TMF763 REST Design Guidelines).
    - Resource version identifiers MAY use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
    - OpenGatewayProductOffering resource SHALL be able to be versioned in the catalog.
    - OpenGatewayProductOffering resource SHALL be able to reference an ApiProductSpecification by id and version.
    - An API Version addition or deletion, or status update to an ApiProductSpecification MAY imply the definition of a new version of that specification.
    - A new ApiProductSpecification MAY imply the definition of a new version of OpenGatewayProductOffering referencing it by id and version.
- Therefore, the TMF931 Onboarding Operate API MAY use this TMF936 catalog API to:
    - Provision OpenGatewayProductOfferings based on its ApiProductSpecification, including the available API versions for each ApiProduct.
    - Manage the configuration of available ApiProducts to Applications and updating the availability of a set of API versions when the ApiProductSpecification is updated. Note that this management can also be done by the Operator itself without use of TMF931.
    - Inventory ApiProduct SHALL be able to reference its OpenGatewayProductOffering and ApiProductSpecification by id and version.

Schemas lifecycleStatus

The lifecycleStatus proposal comes from TMF620 Product Catalog v5 API User Guide.

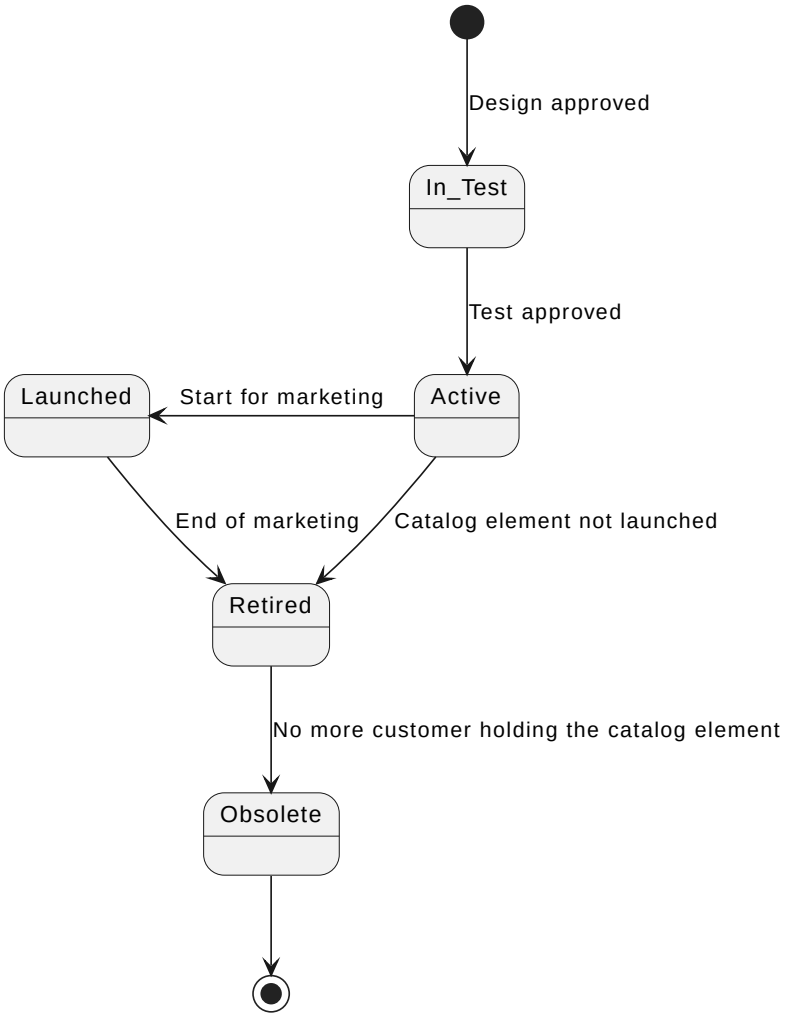
This context restriction is that lifecycleStatus "launched" is proposed only for ProductOffering. "launched" attribute is not needed for other schemas as only the ProductOffering is really launched by marketing team.

For same ProductOffering : only 1 version should have lifecycleStatus at "launched". For other schemas : only 1 version should have lifecycleStatus at "active". The response on the GET /entity MUST answer with the latest version of the entity.

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For a productOffering with "launched" lifecycleStatus, all the other sub resource ProductSpecification, ProductOfferingPrice and ProductOfferingTermOrCondition should have at least 1 instance with "active" lifecycleStatus (if sub resource presence is relevant).

ProductOffering lifecycleStatus



When the test is OK then its status is changed to “active”. When a catalog element is in a “Active” status it means, it has been validated and tested, but it is still not available for customers.

When the beginning of marketing is reached, its status is changed to “launched”. At this moment, customers can buy it.

If the catalog element is not launched, its status is changed to “retired”.

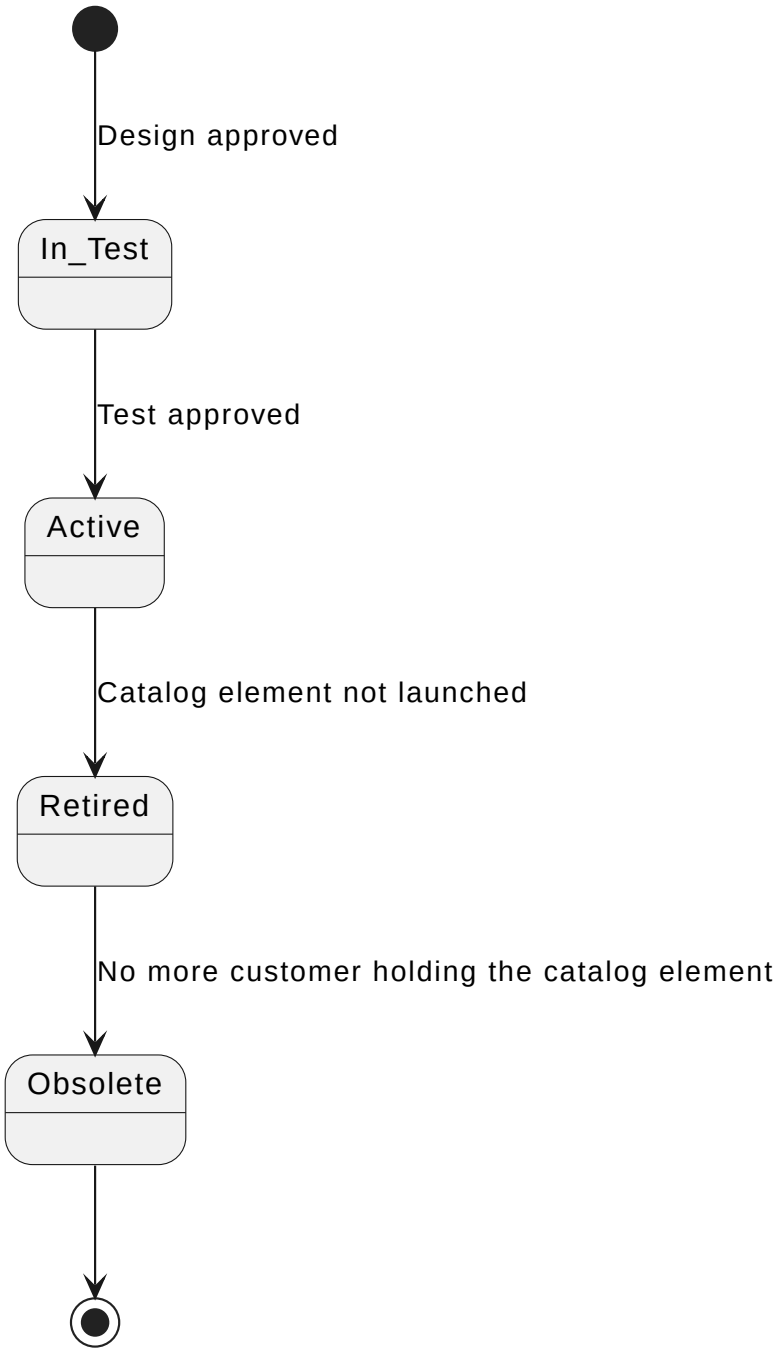
The same status is achieved when a catalog element reaches the end of marketing.

The “retired” status means it cannot be sold to any new customers, but previous customers can still have it.

When no more customer holds the catalog element, its status is changed to “obsolete” meaning it can be removed from the catalog.

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ProductSpecification and ProductOfferingPrice lifecycleStatus



ProductOffering and ProductSpecification version

These schemas MAY use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix. Catalog entity versioning follows the Versioned Entity Model Pattern from TMF763 REST Design Guidelines v5.0.0.

ProductOffering and ProductSpecification may be referenced by `id` and `version`. For example, an ApiProductOffering may be ordered at TMF931 Open Gateway Onboarding and Ordering API by requesting a particular version of the offering provided for its catalogue. But an OpenGatewayProductOffering should be ordered at TMF931 Open Gateway Onboarding

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and Ordering API only by requesting only its id to order the current version. This current version of a ProductOffering exposed by the Operator should have a lifecycleStatus "launched" to allow it to be ordered. Indeed, an OpenGatewayProductOffering may also reference an OpenGatewayProductOffering by `id` and `version`.

```
{
    "@type": "ProductOfferingRef",
    "@referredType": "OpenGatewayProductOffering",
    "id": "233434444434353",
    "version": "1.2.0"
}
```

Version Based Filtering

For Query entities Operation, The version selector keyword can be used in the query string, where the value of the keyword is the semantic version, version=major.minor.patch.

For example: version=1.1.0. Only the equals operator = is supported for the version selector. Other operators are not supported.

- If version selector is set to a specific version, the operation applies to that version of the entity.
- If version selector is not present, the operation applies to the current version of the entity.
- If version selector is set to the reserved value all (valid only for GET on the entity collection), the GET operation applies to all versions of the entity.

This applies only when version is specified directly in the query string. If version is specified within a filter or json-path parameter, it will be treated like any other expression and will be applied to the version property of the entity.

Entity versions can be retrieved using the `version` query param with the equal `=` operator and will return an ApiProductOffering if it matches with the `version` property of the stored entity version.

For example, an OpenGatewayProductOffering version may be retrieved with next operation

```
GET /productOffering/233434444434353?version=1.2.0
```

```
{
    "id": "233434444434353",
    "href":
"/productcatalogManagement/productOffering/233434444434353?version=1.2.0",
    "@type": "OpenGatewayProductOffering",
    "@baseType": "DcsProductOffering",
    "version": "1.2.0",
    "versionMetadata": {
        "id": "1123347354",
        "href":
"/productCatalogManagement/productOffering/233434444434353/version/1123347354",
        "@type": "Version",
        "version": "1.0",
```

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```
        "isCurrent": false,
        "creationDate": "2013-04-19T16:42:23-04:00",
        "@versionedEntityType": "OpenGatewayProductOffering",
        "versionedBy": {
            "role": "catalogAdmin",
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "partyOrPartyRole": {
                "id": "EMP443462357342351254562 52",
                "href": "/partyRoleManagement/partyRole/EMP443462357342351254562 52",
                "name": "Joan Doe",
                "partyID": "PTY3546233462533463445624 52",
                "partyName": "Joan Doe",
                "@type": "PartyRoleRef",
                "@referredType": "Employee"
            }
        }
    },
    "lastUpdate": "2013-04-19T16:42:23-04:00",
    "name": "Virtual Storage Medium",
    "description": "Virtual Storage Medium",
    "isBundle": "true",
    "lifecycleStatus": "Inactive"
}
```


- Retrieve a specific version of a versioned entity

When the regular GET retrieve endpoint is invoked on a versioned entity, the current version is returned.

- Querying versioned entities

When the GET list/search endpoint (on the entity collection) is invoked on a versioned entity, and the version selector is not specified as part of the filter expression, the result set will return only current versions of entities that match the filter

- Query all version metadata for an entity

If there is no version number mentioned, it returns the meta data for all the versions of specific product offering. The query looks like this:

```
GET <basepath>/<api>/<entity>/<id>/version
```

Strongly typed schemas

Impact on Product Ordering and Product Inventory ^^^^

A Product Ordering business process requires the execution of a series of previous actions (including API calls) in a defined order so that the process can be carried out correctly. Like the more generic product ordering process managed with TMF Product Ordering, the specialized API Product Ordering is also catalog-oriented, so in order to be carried out it needs to know, through the execution of previous steps, the specifications of the products to be delivered in the order, as well as the actions required by the requester on these products. There is a clear alignment between the way a product is modelled in the product catalog and the way this product can be ordered via an API Product Order call. The modelling of the

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productSpecification or ProductOffering in the Product Catalog has also an impact on the way the Product is finally reflected in the product inventory.

With the specialized ProductOrder resource definition in TMF931 Onboarding and Ordering as a specialization of a ProductOrder resource, a new way to ensure such alignment has been introduced: the use of strongly typed referenced schemas for ProductOrderItem and ProductAction. The underlying idea can be summarized as follows: each ProductSpecification managed in a ProductOrder requires a set of optional or mandatory attributes to be valued by the API client to ensure its successful delivery. The attributes to be in input depend on the productSpecification definition and the action to be performed over it (add, modify or delete), as each product may require different information. A schema-based model is mandatory : there are for each ProductSpecification definition (in this context passed at the atomic ProductOffering level) a reference to N ProductAction schemas, one for each possible action available over that ProductSpecification (via a ProductOffering). Each schema includes the corresponding set of attributes necessary for that ProductOffering/ProductSpecification to be delivered via a ProductOrder. Same applies for the Product : for each ProductSpecification/ProductOffering there is a reference to a Product schema that includes the corresponding set of attributes necessary for that Product description.

Examples

Let's take the example of a Location Verification API that is offerd through an atomic ProductOffering based on a ProductSpecification. Let's assume that this ProductSpecification describes a specific characteristic for this API (coming from the Customer Facing Service definition) : bestAccuracy. This characteristic is not in the common schemas Product and ProductSpecification, so it's not described in the OAS, because the OAS can't contain all specific schemas from all product specifications that will be exposed in all Operator catalogs. This rule permits to not having to upgrade the Operate API OAS each time a new ProductSpecification is added into an Operator catalog.

We need to use specialized schemas that will allow this specific characteristic to be valued and read by the Operate API client. So it has to be described in a JSON schema that the API client can discover. The way TMF decided to do is :

- @type : name of the schema. It can be the standard name of schema from OAS, or can be a specific name because the schema needs to be specialized
- @schemaLocation : URL of the json schema describing the schema used in the payload. This attribute is not mandatory when @type value is the standard OAS schema; but becomes mandatory when the @type is the name of a schema not in the OAS.
- @baseType : can be used when the schema is specialized in order to indicate the parent schema

Call flows

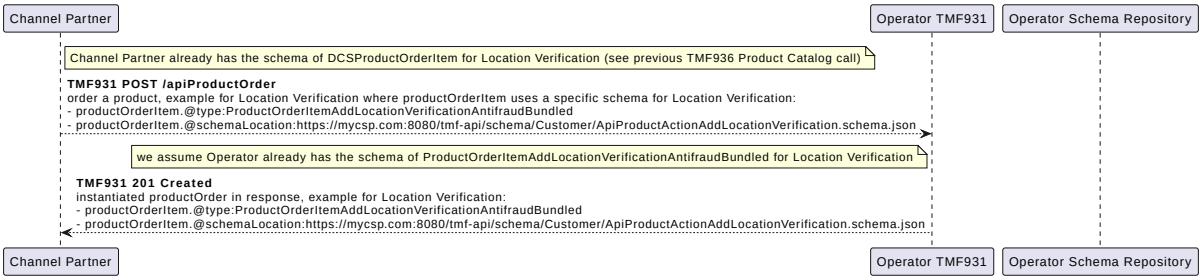
In order to better understand how to manage strongly typed shemas, this call flow is an illustrative way of doing it.

Getting product offering and product specification

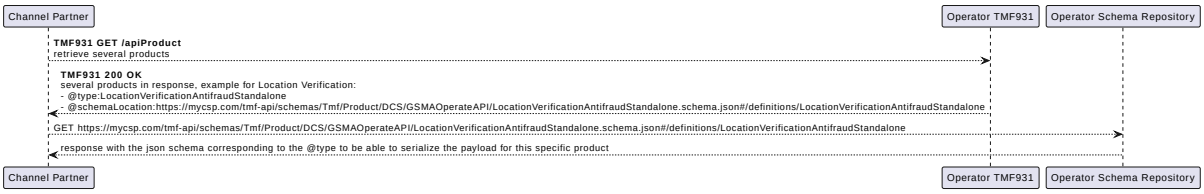
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Posting product order use case



Reading product inventory use case



Note that the folder structure for json schema in the examples is not normative.

json schemas repository

Each Operator maintains its own repository of json schemas that contain the definitions of their offers with their characteristics and possible values. These schemas are a strict specialization of standard json schemas maintained under GSMA in this repository : https://github.com/GSMA-Open-Gateway/Operate-API-References. So the name of attributes, their types, the possible values, are common to all Operators. But each Operator can limits the possible values or the possible attributes present in a product Specification (targetProductSchema).

String property limits

As most of the string properties in Operate APIs have no limit and it may cause interoperability issues, here are some guidelines depending on the property type:

Property Type	Max. Length (chars)	Format	
URL / URI	2048 (common browsers' max length)	URI/URL format as defined in [RFC3986](https://datatracker.ietf.org/doc/html/rfc3986)	

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id	200		
description	1000		
other	200		

> **NOTE**: Default JSON string encoding UTF-8 is assumed. UUID format is recommended for ids: 32 char String format defined in [RFC4122](https://datatracker.ietf.org/doc/html/rfc4122). For example: `f81d4fae-7dec-11d0-a765-00a0c91e6bf6`

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a ApiProductSpecification inheriting properties from the OpenGatewayProductSpecification entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of OpenGatewayProductSpecification instances some may be instances of ApiProductSpecification where other could be instances of UsageVolumeProductSpecification (not present in this specification version). The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance a ProductSpecificationRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as ApiProductSpecificationRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

Note : some schemas have been added to the specification only for giving examples of specialized schemas for a particular product : ApiProductActionAddDeviceLocationVerification, ApiProductActionAddDeviceLocationVerificationBundled, ApiProductActionModifyDeviceLocationVerification, ApiProductDeviceLocationVerification, ApiProductLocationVerificationAntifraud, UsageVolumeProductLocationVerification. They should not be implemented as is.

OpenGatewayProductOffering resource

The specialized ProductOffering for GSMA Open Gateway Operate API.

Resource model

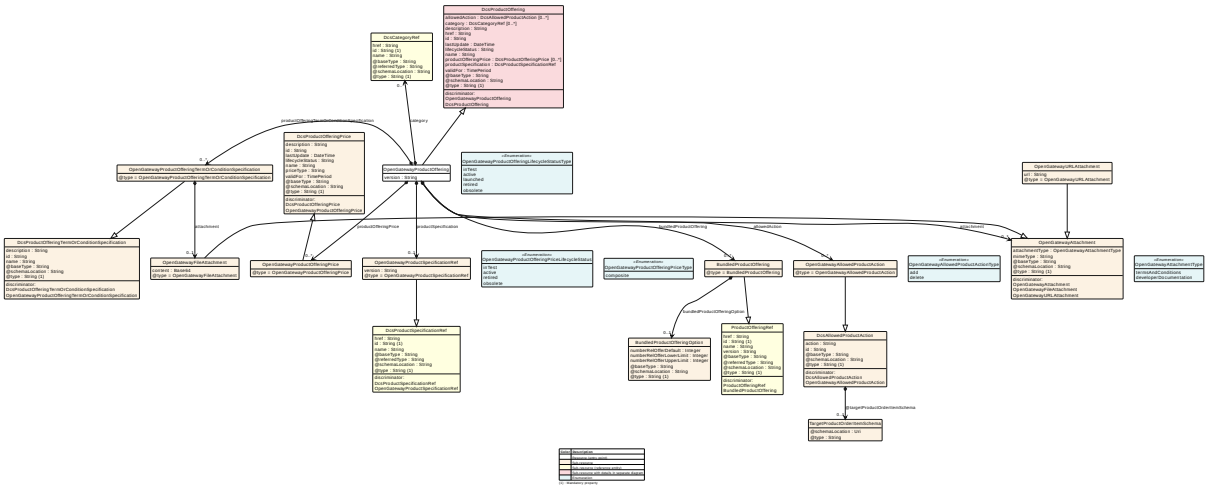


Figure 1 - OpenGatewayProductOffering

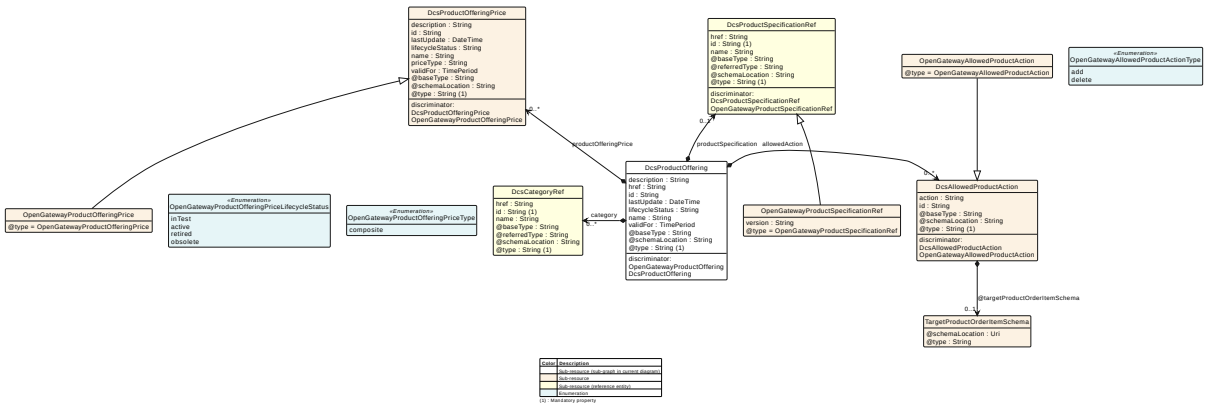


Figure 2 - DcsProductOffering

Field descriptions

OpenGatewayProductOffering fields

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allowedAction	An OpenGatewayAllowedProductAction. List of actions that can be executed (in context of a product order) on products instantiated from this offering.
attachment	An OpenGatewayAttachment. This attribute can be used to attach additional documentation for the productOffering, like marketing documentation … OpenGatewayAttachment can be instantiated as * OpenGatewayFileAttachment * OpenGatewayURLAttachment
bundledProductOffering	A BundledProductOffering. A type of ProductOffering that belongs to a grouping of ProductOfferings made available to the market. It inherits of all attributes of ProductOffering.
category	A DcsCategoryRef. NOT USED YET IN OPEN GATEWAY CONTEXT.
description	A String. Description of the productOffering.
href	A String. Hyperlink reference.
id	A String. Unique identifier. Recommended UUID format.
lastUpdate	A DateTime. Date and time of the last update.
lifecycleStatus	An OpenGatewayProductOfferingLifecycleStatusType. Used to indicate the ProductOffering lifecycleStatus with limited enum values for Open Gateway context. ENUMERATED with values: * inTest * active * launched * retired * obsolete
name	A String. Name of the productOffering.
productOfferingPrice	An OpenGatewayProductOfferingPrice. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased. The price is valid for a defined period of time and may not represent the actual price paid by a customer.
productOfferingTermOrConditionSpecification	An OpenGatewayProductOfferingTermOrConditionSpecification. List of ProductOfferingTermOrConditionSpecification applicable for the ProductOffering.
productSpecification	An OpenGatewayProductSpecificationRef. The specialized ProductSpecification for GSMA Open Gateway Operate API.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the ProductOffering. Version SHOULD use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

OpenGatewayAllowedProductAction sub-resource fields

action	An OpenGatewayAllowedProductActionType. Used to indicate the AllowedProductAction action with limited enum values for Open Gateway context. ENUMERATED with values: * add * delete
id	A String. Unique identifier of the AllowedProductAction within the array.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@targetProductOrderItemSchema	A TargetProductOrderItemSchema. The reference object to the schema and type of target product order item which is described by product allowed action.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductOrderItemSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product order item.
@type	A String. Class type of the target product order item.

DcsAllowedProductAction sub-resource fields

action	A String. The name of the action. Values must be taken from this list (or extended it adding more values) : add, modify, delete.
id	A String. Unique identifier of the AllowedProductAction within the array.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@targetProductOrderItemSchema	A TargetProductOrderItemSchema. The reference object to the schema and type of target product order item which is described by product allowed action.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DcsAllowedProductAction can be instantiated as * OpenGatewayAllowedProductAction

OpenGatewayAttachment sub-resource fields

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attachmentType	An OpenGatewayAttachmentType. Used to indicate the Attachment attachmentType with limited enum values for Open Gateway context. ENUMERATED with values: * termsAndConditions * developerDocumentation
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	OpenGatewayAttachment can be instantiated as * OpenGatewayFileAttachment * OpenGatewayURLAttachment
content	This property is present in subclasses
url	This property is present in subclasses

OpenGatewayFileAttachment sub-resource fields

attachmentType	An OpenGatewayAttachmentType. Used to indicate the Attachment attachmentType with limited enum values for Open Gateway context. ENUMERATED with values: * termsAndConditions * developerDocumentation
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A String. Describes the possible values for the mimeType of the attachment. A media type (also known as a Multipurpose Internet Mail Extensions or MIME type) indicates the nature and format of a document, file, or assortment of bytes. MIME types are defined and standardized in IETF's RFC 6838. ENUMERATED with values: * application/pdf
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OpenGatewayURLAttachment sub-resource fields

attachmentType	An OpenGatewayAttachmentType. Used to indicate the Attachment attachmentType with limited enum values for Open Gateway context. ENUMERATED with values: * termsAndConditions * developerDocumentation

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mimeType	A String. Describes the possible values for the mimeType of the attachment. A media type (also known as a Multipurpose Internet Mail Extensions or MIME type) indicates the nature and format of a document, file, or assortment of bytes. MIME types are defined and standardized in IETF's RFC 6838.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BundledProductOffering sub-resource fields

bundledProductOfferingOption	A BundledProductOfferingOption. A set of numbers that specifies the lower and upper limits for a ProductOffering that can be procured as part of the related BundledProductOffering. Values can range from 0 to unbounded.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BundledProductOfferingOption sub-resource fields

numberRelOfferDefault	An Integer. Default number of produc offereings that should be procured as part of the related BundledProductOffering.
numberRelOfferLowerLimit	An Integer. Lower limit for a product offering that can be procured as part of the related BundledProductOffering.
numberRelOfferUpperLimit	An Integer. Upper limit for a product offering that can be procured as part of the related BundledProductOffering.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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ProductOfferingRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ProductOfferingRef can be instantiated as * BundledProductOffering
bundledProductOfferingOption	This property is present in subclasses

DcsCategoryRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OpenGatewayProductOfferingPrice sub-resource fields

description	A String. Description of the productOfferingPrice.
id	A String. Unique identifier of the ProductOfferingPrice within the array.
lastUpdate	A DateTime. The last update time of this ProductOfferingPrice.

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lifecycleStatus	An OpenGatewayProductOfferingPriceLifecycleStatus. Used to indicate the current ProductOfferingPrice lifecyclestatus with limited enum values for Open Gateway context. ENUMERATED with values: * inTest * active * retired * obsolete
name	A String. Name of the productOfferingPrice.
priceType	An OpenGatewayProductOfferingPriceType. Used to indicate the ProductOfferingPrice priceType with limited enum values for Open Gateway context. Kept for base schema ProductOfferingPrice backward compatibility reason. ENUMERATED with values: * composite
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DcsProductOfferingPrice sub-resource fields

description	A String. Description of the productOfferingPrice.
id	A String. Unique identifier of the ProductOfferingPrice within the array.
lastUpdate	A DateTime. The last update time of this ProductOfferingPrice.
lifecycleStatus	A String. Used to indicate the current lifecycle status. Values must be taken from this list (or extended it adding more values for a specific context) : inStudy, inDesign, inTest, active, launched, retired, obsolete, rejected.
name	A String. Name of the productOfferingPrice.
priceType	A String. Used to indicate the current price type. Values must be taken from this list (or extended it adding more values for a specific context) : usage, oneTime, recurring, discountFee, discountPercentage, taxFee, taxPercentage, taxRule, replacementUsage, remplacementOneTime, replacementRecurring.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DcsProductOfferingPrice can be instantiated as * OpenGatewayProductOfferingPrice

OpenGatewayProductOfferingTermOrConditionSpecification sub-resource fields

attachment	An OpenGatewayFileAttachment. Binary file attachment.
description	A String. A narrative that explains in detail what the productOfferingTermOrConditionSpecification is.
id	A String. Locally unique identifier of the productOfferingTermOrConditionSpecification within its product offering.
name	A String. Name of the productOfferingTermOrConditionSpecification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DcsProductOfferingTermOrConditionSpecification sub-resource fields

description	A String. A narrative that explains in detail what the productOfferingTermOrConditionSpecification is.
id	A String. Locally unique identifier of the productOfferingTermOrConditionSpecification within its product offering.
name	A String. Name of the productOfferingTermOrConditionSpecification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DcsProductOfferingTermOrConditionSpecification can be instantiated as * OpenGatewayProductOfferingTermOrConditionSpecification
attachment	This property is present in subclasses

OpenGatewayProductSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
version	A String. Version of the ProductSpecification. Version SHOULD use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DcsProductSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DcsProductSpecificationRef can be instantiated as * OpenGatewayProductSpecificationRef
version	This property is present in subclasses

Json representation sample(s)

We provide below a JSON representation as example of the 'OpenGatewayProductOffering' resource object.


```
{
  "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
  "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
  "lifecycleStatus": "launched",
  "version": "1.0.0",
  "name": "Location Verification Antifraud CAMARA API Standalone",
  "description": "Location Verification product offering standalone, for antifraud
purpose, accessible with CAMARA API.",
  "lastUpdate": "2023-07-01T00:00:00Z",
  "allowedAction": [
```


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```
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "1",
      "action": "add",
      "@targetProductOrderItemSchema": {
        "@type": "ProductOrderItemAddLocationVerificationAntifraudStandalone",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ProductOrderItemAddLocationVerificationAntifraudStandalone"
      }
    },
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "2",
      "action": "delete",
      "@targetProductOrderItemSchema": {
        "@type": "DcsProductOrderItemDelete",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
      }
    }
  ],
  "productSpecification": {
    "@type": "OpenGatewayProductSpecificationRef",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
  },
  "productOfferingTermOrConditionSpecification": [{
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
    "name": "Location Verification Antifraud CAMARA API Standalone Product Offering
Terms and Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
      "@type": "OpenGatewayFileAttachment",
      "content": "JVBERi0xL...DQolJUVPRg==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }],
  "productOfferingPrice": [
    {
      "id": "1",
      "@type": "OpenGatewayProductOfferingPrice",
      "lifecycleStatus": "active",
      "name": "Location Verification Antifraud CAMARA API Standalone Usage Price",
      "description": "Location Verification for Antifraud standalone product
offering price, usage price. 9 EUR per API call for usage verify for FR, ES and GB.
9 USD per API call for usage verify for US.",
      "lastUpdate": "2023-09-01T00:00:00Z",
      "priceType": "composite"
    }
  ]
}
```


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OpenGatewayProductSpecification resource

The specialized ProductSpecification for GSMA Open Gateway Operate API.

Resource model

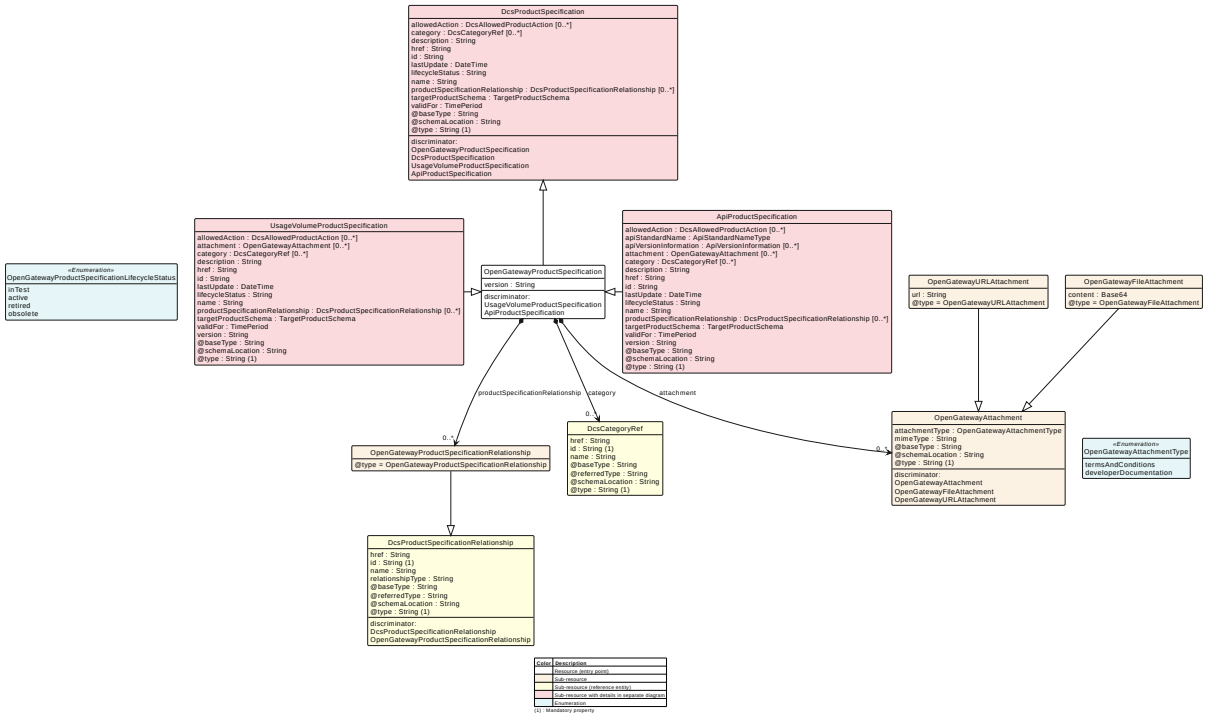


Figure 3 - OpenGatewayProductSpecification

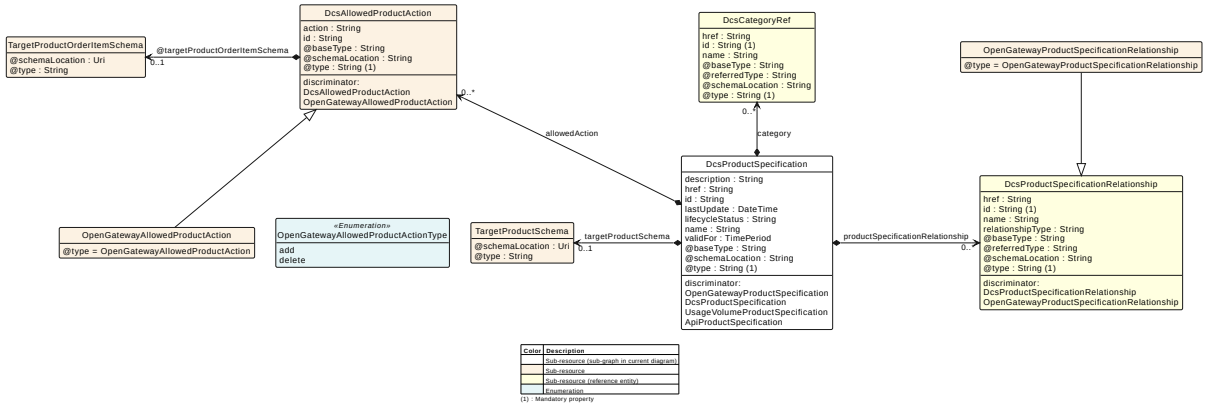


Figure 4 - DcsProductSpecification

Field descriptions

OpenGatewayProductSpecification fields

allowedAction	A DcsAllowedProductAction. List of actions that can be executed (in context of a product order) on products instantiated from this productSpecification. DcsAllowedProductAction can be instantiated as * OpenGatewayAllowedProductAction

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attachment	An OpenGatewayAttachment. This attribute can be used to attach additional documentation for the productSpecification, like technical documentation, frequently asked questions, getting started, … OpenGatewayAttachment can be instantiated as * OpenGatewayFileAttachment * OpenGatewayURLAttachment
category	A DcsCategoryRef. NOT USED YET IN OPEN GATEWAY CONTEXT.
description	A String. Description of the productSpecification.
href	A String. Hyperlink reference.
id	A String. Unique identifier. Recommended UUID format.
lastUpdate	A DateTime. Date and time of the last update.
lifecycleStatus	An OpenGatewayProductSpecificationLifecycleStatus. Used to indicate the current ProductSpecification lifecycleStatus with limited enum values for Open Gateway context. ENUMERATED with values: * inTest * active * retired * obsolete
name	A String. Name of the productSpecification.
productSpecificationRelationship	An OpenGatewayProductSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among product specifications.
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the ProductSpecification. Version SHOULD use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	OpenGatewayProductSpecification can be instantiated as * ApiProductSpecification * UsageVolumeProductSpecification
apiStandardName	This property is present in subclasses
apiVersionInformation	This property is present in subclasses

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DcsAllowedProductAction sub-resource fields

See [DcsAllowedProductAction] for details.

TargetProductOrderItemSchema sub-resource fields

See [TargetProductOrderItemSchema] for details.

OpenGatewayAllowedProductAction sub-resource fields

See [OpenGatewayAllowedProductAction] for details.

OpenGatewayAttachment sub-resource fields

See [OpenGatewayAttachment] for details.

OpenGatewayFileAttachment sub-resource fields

See [OpenGatewayFileAttachment] for details.

OpenGatewayURLAttachment sub-resource fields

See [OpenGatewayURLAttachment] for details.

DcsCategoryRef sub-resource fields

See [DcsCategoryRef] for details.

OpenGatewayProductSpecificationRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Indicates the possible values of relationshipType between ProductSpecification. requires means that the related productSpecification is required to be ordered before; appliesOne means that the related productSpecification describes the product to which the productspecification will apply (for example a UsageVolume product applies on an Api product). ENUMERATED with values: * requires * appliesOn
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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DcsProductSpecificationRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of the relationship. Values must be taken from this list (or extended it adding more values for a specific context) : requires, appliesOn.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DcsProductSpecificationRelationship can be instantiated as * OpenGatewayProductSpecificationRelationship

TargetProductSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product.
@type	A String. Class type of the target product.

ApiVersionInformation sub-resource fields

apiBasePath	A String. Base path of the API. CAMARA project recommends to set the base path as `/<apiName>/v<major>` or `/apiName/v0.<minor>` (minor version only allowed for unstable APIs in 0 major version).
apiGrantInformation	An ApiGrantInformation. ApiGrantInformation is a structured set of information regarding the grant of access to an API product, detailing its purpose, scope, grant type, and legal basis for usage.
apiName	A String. Exact name of API. For CAMARA standard it must correspond to the API name column of this table https://github.com/camaraproject. Example : location-verification. It may happen that the exact name of the API changes from a version to another.
apiStatus	A String. Is the lifecycle status of the API. Active means the Application can consume this API on this version after a future order. Retired means that the Application can consume this API on this version only if the order has already been done, no new order on this Product Specification can give the right to consume this API on this version. ENUMERATED with values: * active * retired

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apiVersion	A String. Version of the API. CAMARA project recommends [Semantic Versioning 2.0.0](https://semver.org/) where versions are definied as x.y.z version number which corresponds to a major, minor and fix.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiGrantInformation sub-resource fields

grantType	A String. One or several grantType of the API, as stated in the API that has been purchased. Examples: "authorization_code" "client_credentials" "urn:ietf:params:oauth:grant-type:token-exchange" "urn:ietf:params:oauth:grant-type:jwt-bearer" "urn:ietf:params:oauth:grant-type:saml2-bearer" "urn:ietf:params:oauth:grant-type:device_code" "urn:openid:params:grant-type:ciba".
legalBasis	A DpvLegalBasisType. Valid values for Legal Basis based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:Consent * dpv:Contract * dpv:ContractPerformance * dpv:DataTransferLegalBasis * dpv:EnterIntoContract * dpv:LegalBasis * dpv:LegalObligation * dpv:LegitimateInterest * dpv:LegitimateInterestOfController * dpv:LegitimateInterestOfDataSubject * dpv:LegitimateInterestOfThirdParty * dpv:OfficialAuthorityOfController * dpv:PublicInterest * dpv:VitalInterest * dpv:VitalInterestOfDataSubject * dpv:VitalInterestOfNaturalPerson * dpv:InformedConsent * dpv:DataControllerContract * dpv:ImpliedConsent * dpv:DataSubjectContract * dpv:ExplicitlyExpressedConsent * dpv:UninformedConsent * dpv:DataProcessorContract * dpv:ExpressedConsent * dpv:ThirdPartyContract

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95

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scope	A String. One or several scopes of the API, as stated in the API that has been purchased. Examples: "sim-swap:retrieve-date".
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'OpenGatewayProductSpecification' resource object.


```
{
  "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
  "name": "Location Verification Antifraud CAMARA API",
  "version": "1.0.0",
  "@type": "ApiProductSpecification",
        "@baseType": "OpenGatewayProductSpecification",
  "lifecycleStatus": "active",
  "description": "Location Verification product specification, for antifraud
purpose, accessible with CAMARA API.",
  "lastUpdate": "2023-09-23T16:42:23Z",
  "targetProductSchema": {
    "@type": "ApiProductLocationVerificationAntifraud",
    "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/LocationVerificationAntifraudStandalone"
  },
  "apiStandardName": "CAMARA",
  "apiVersionInformation": [
    {
      "@type": "ApiVersionInformation",
      "apiName": "location-verification",
      "apiVersion": "1.0.0",
      "apiBasePath": "/location-verification/v1",
      "apiStatus": "active",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "legalBasis": "dpv:Consent",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "authorization_code",
            "urn:openid:params:grant-type:ciba"
          ]
        }
      ]
    }
  ],
  "attachment": [
```


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```
    {
      "@type": "OpenGatewayURLAttachment",
      "url":
"https://mycsp.com/OpenGateway/documentation/LocationVerificationDeveloperGuide.html
",
      "attachmentType": "developerDocumentation",
      "mimeType": "text/html"
    }
  ]
}
```

ApiProductSpecification resource

Specification for the API Product with the supported API versions.

Resource model

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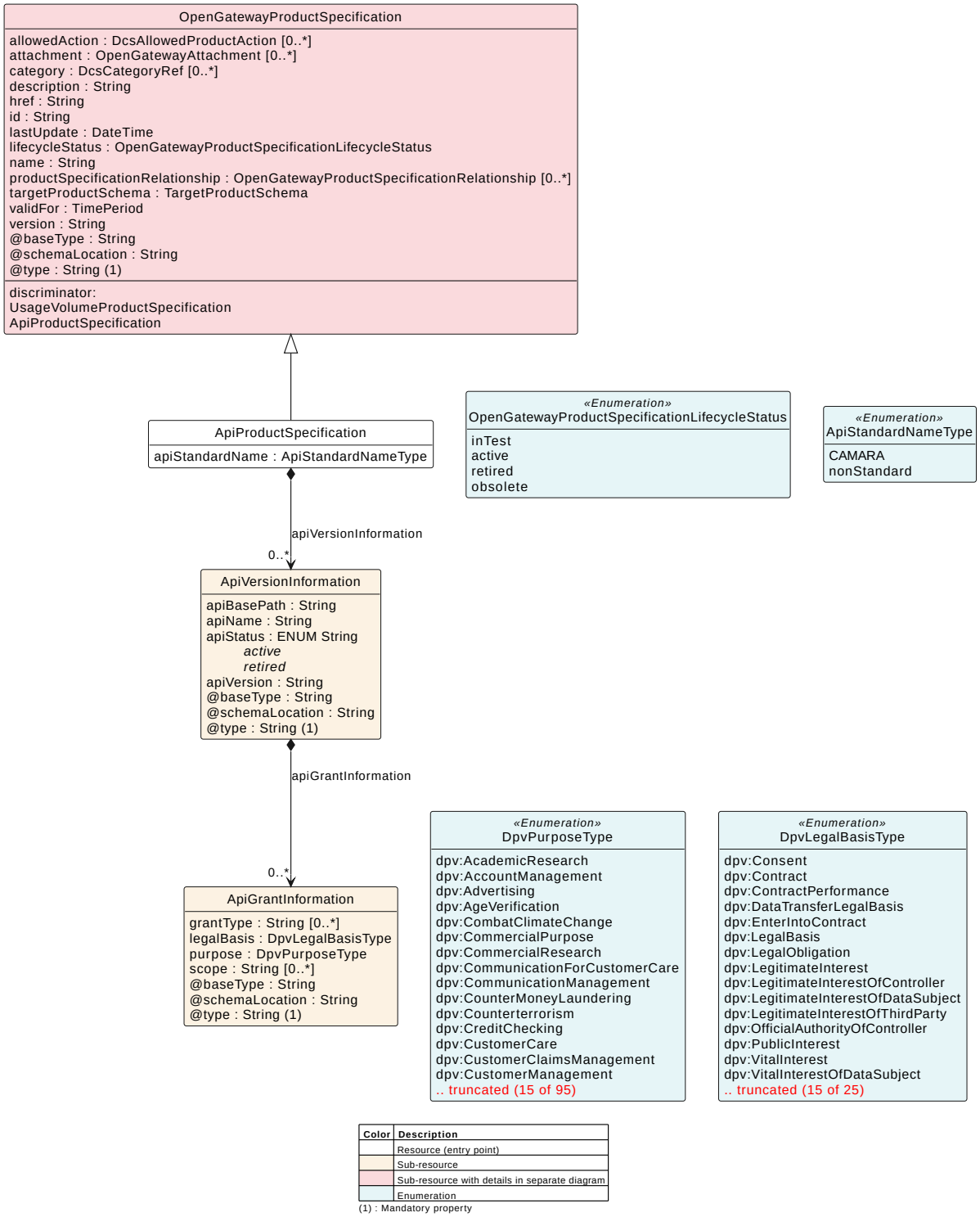


Figure 5 - ApiProductSpecification

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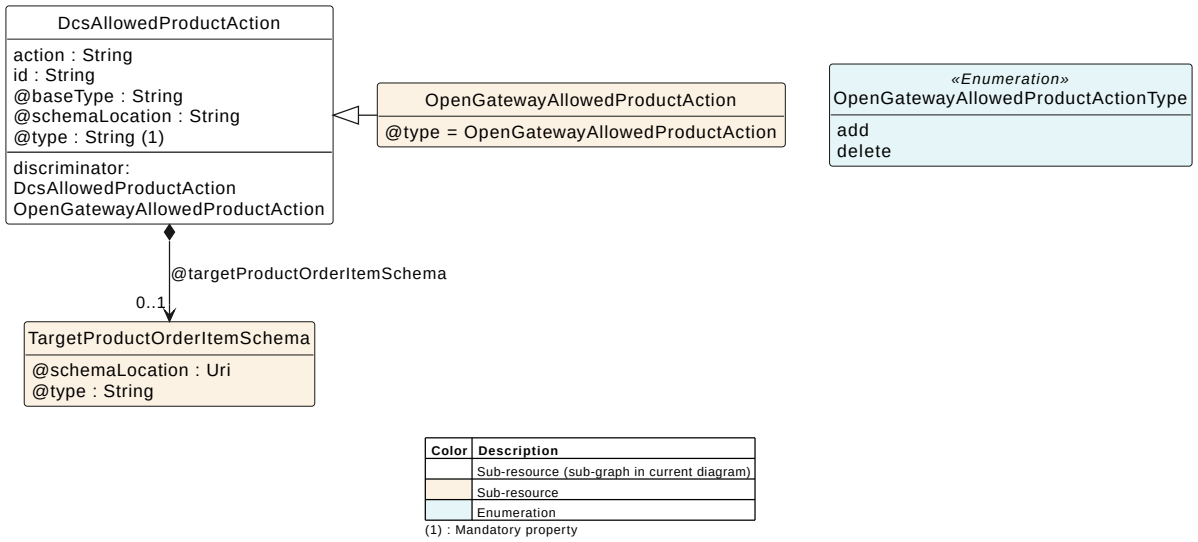


Figure 6 - DcsAllowedProductAction

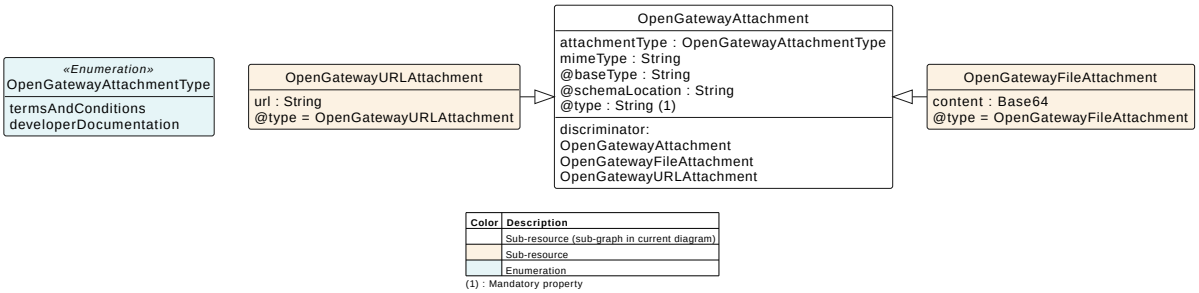


Figure 7 - OpenGatewayAttachment

See Figure 4 for DcsProductSpecification diagrams.

Field descriptions

ApiProductSpecification fields

allowedAction	A DcsAllowedProductAction. List of actions that can be executed (in context of a product order) on products instantiated from this productSpecification. DcsAllowedProductAction can be instantiated as * OpenGatewayAllowedProductAction
apiStandardName	An ApiStandardNameType. Valid values for the name of standard organization of project that is responsible of the API specification. ENUMERATED with values: * CAMARA * nonStandard
apiVersionInformation	An ApiVersionInformation. All values of apiVersionInformation in a same ApiProductSpecification should correspond to the same API.

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attachment	An OpenGatewayAttachment. This attribute can be used to attach additional documentation for the productSpecification, like technical documentation, frequently asked questions, getting started, … OpenGatewayAttachment can be instantiated as * OpenGatewayFileAttachment * OpenGatewayURLAttachment
category	A DcsCategoryRef. The category resource is used to group product specifications in logical containers. Categories can contain other categories and/or other catalog entries.
description	A String. A narrative that explains in detail what the product specification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time of the last update.
lifecycleStatus	An OpenGatewayProductSpecificationLifecycleStatus. Used to indicate the current ProductSpecification lifecycleStatus with limited enum values for Open Gateway context. ENUMERATED with values: * inTest * active * retired * obsolete
name	A String. Name of the product specification.
productSpecificationRelationship	A DcsProductSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among product specifications. DcsProductSpecificationRelationship can be instantiated as * OpenGatewayProductSpecificationRelationship
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the ProductSpecification. Version SHOULD use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DcsAllowedProductAction sub-resource fields

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See [DcsAllowedProductAction] for details.

TargetProductOrderItemSchema sub-resource fields

See [TargetProductOrderItemSchema] for details.

OpenGatewayAllowedProductAction sub-resource fields

See [OpenGatewayAllowedProductAction] for details.

ApiVersionInformation sub-resource fields

See [ApiVersionInformation] for details.

ApiGrantInformation sub-resource fields

See [ApiGrantInformation] for details.

OpenGatewayAttachment sub-resource fields

See [OpenGatewayAttachment] for details.

OpenGatewayFileAttachment sub-resource fields

See [OpenGatewayFileAttachment] for details.

OpenGatewayURLAttachment sub-resource fields

See [OpenGatewayURLAttachment] for details.

DcsCategoryRef sub-resource fields

See [DcsCategoryRef] for details.

DcsProductSpecificationRelationship sub-resource fields

See [DcsProductSpecificationRelationship] for details.

OpenGatewayProductSpecificationRelationship sub-resource fields

See [OpenGatewayProductSpecificationRelationship] for details.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApiProductSpecification' resource object.

```
{
  "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
  "href": "https://mycsp.com/tmf-
```

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```
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
  "name": "Location Verification Antifraud CAMARA API",
  "version": "1.0.0",
  "@type": "ApiProductSpecification",
        "@baseType": "OpenGatewayProductSpecification",
  "lifecycleStatus": "active",
  "description": "Location Verification product specification, for antifraud
purpose, accessible with CAMARA API.",
  "lastUpdate": "2023-09-23T16:42:23Z",
  "targetProductSchema": {
    "@type": "ApiProductLocationVerificationAntifraud",
    "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/LocationVerificationAntifraudStandalone"
  },
  "apiStandardName": "CAMARA",
  "apiVersionInformation": [
    {
      "@type": "ApiVersionInformation",
      "apiName": "location-verification",
      "apiVersion": "1.0.0",
      "apiBasePath": "/location-verification/v1",
      "apiStatus": "active",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "legalBasis": "dpv:Consent",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "authorization_code",
            "urn:openid:params:grant-type:ciba"
          ]
        }
      ]
    }
  ],
  "attachment": [
    {
      "@type": "OpenGatewayURLAttachment",
      "url":
"https://mycsp.com/OpenGateway/documentation/LocationVerificationDeveloperGuide.html
",
      "attachmentType": "developerDocumentation",
      "mimeType": "text/html"
    }
  ]
}
```


UsageVolumeProductSpecification resource

Specification for the Usage Volume Product.

Resource model

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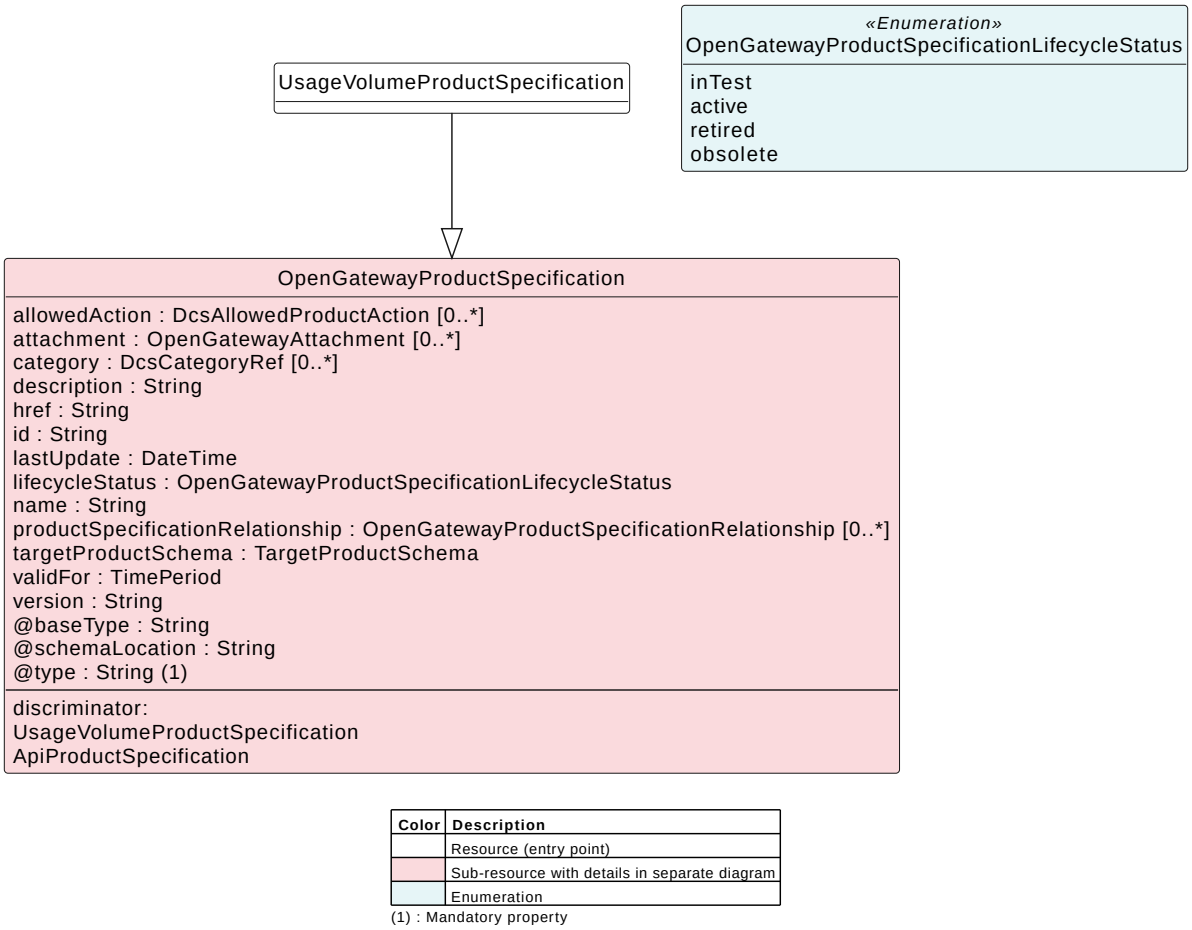


Figure 8 - UsageVolumeProductSpecification

See Figure 6 for DcsAllowedProductAction diagrams.

See Figure 7 for OpenGatewayAttachment diagrams.

See Figure 4 for DcsProductSpecification diagrams.

Field descriptions

UsageVolumeProductSpecification fields

allowedAction	A DcsAllowedProductAction. List of actions that can be executed (in context of a product order) on products instantiated from this productSpecification. DcsAllowedProductAction can be instantiated as * OpenGatewayAllowedProductAction

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attachment	An OpenGatewayAttachment. This attribute can be used to attach additional documentation for the productSpecification, like technical documentation, frequently asked questions, getting started, … OpenGatewayAttachment can be instantiated as * OpenGatewayFileAttachment * OpenGatewayURLAttachment
category	A DcsCategoryRef. The category resource is used to group product specifications in logical containers. Categories can contain other categories and/or other catalog entries.
description	A String. A narrative that explains in detail what the product specification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. Date and time of the last update.
lifecycleStatus	An OpenGatewayProductSpecificationLifecycleStatus. Used to indicate the current ProductSpecification lifecycleStatus with limited enum values for Open Gateway context. ENUMERATED with values: * inTest * active * retired * obsolete
name	A String. Name of the product specification.
productSpecificationRelationship	A DcsProductSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among product specifications. DcsProductSpecificationRelationship can be instantiated as * OpenGatewayProductSpecificationRelationship
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the ProductSpecification. Version SHOULD use the semantic versioning X.Y.Z version format where X identifies a major version, Y is the minor version and Z a fix.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DcsAllowedProductAction sub-resource fields

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See [DcsAllowedProductAction] for details.

TargetProductOrderItemSchema sub-resource fields

See [TargetProductOrderItemSchema] for details.

OpenGatewayAllowedProductAction sub-resource fields

See [OpenGatewayAllowedProductAction] for details.

OpenGatewayAttachment sub-resource fields

See [OpenGatewayAttachment] for details.

OpenGatewayFileAttachment sub-resource fields

See [OpenGatewayFileAttachment] for details.

OpenGatewayURLAttachment sub-resource fields

See [OpenGatewayURLAttachment] for details.

DcsCategoryRef sub-resource fields

See [DcsCategoryRef] for details.

DcsProductSpecificationRelationship sub-resource fields

See [DcsProductSpecificationRelationship] for details.

OpenGatewayProductSpecificationRelationship sub-resource fields

See [OpenGatewayProductSpecificationRelationship] for details.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'UsageVolumeProductSpecification' resource object.


```
{
  "@type": "UsageVolumeProductSpecification",
  "@baseType": "OpenGatewayProductSpecification",
  "id": "7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
  "name": "Location Verification Verify Usage Volume",
    "version": "1.0.0",
    "lifecycleStatus": "active",
    "description": "Location Verification Usage Volume of 10000 API calls for verify
location usage, included in a bucket, recurring monthly, usage non blocked when
bucket is empty.",
```


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```
    "lastUpdate": "2023-09-23T16:42:23Z",
    "targetProductSchema": {
      "@type": "UsageVolumeProductLocationVerification",
      "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/UsageVolumeLocationVerification.schema.js
on#/definitions/UsageVolumeProductLocationVerification"
    },
    "productSpecificationRelationship":
    [
      {
        "@type": "OpenGatewayProductSpecificationRelationship",
        "id":"4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "relationshipType": "requires",
        "@referredType":"ApiProductSpecification"
      }
    ]
  }
```

Notification Resource Models

4 notifications are defined for this API.

Notifications related to OpenGatewayProductOffering:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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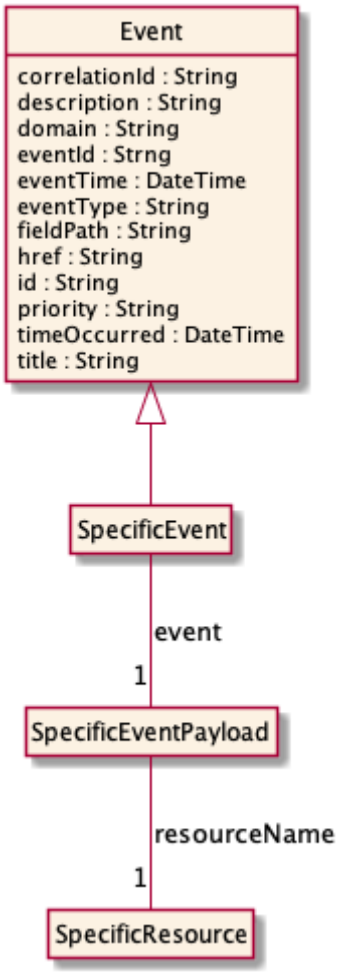


Figure 9 Notification Pattern

OpenGatewayProductOffering

Create Event

Notification for creation of a ProductOffering

```
Content-Type: application/json
{
  "@type": "OpenGatewayProductOfferingCreateEvent",
  "correlationId": "13654384718",
  "domain": "Open Gateway Catalog",
  "timeOccurred": "2024-11-25T14:04:27Z",
  "eventId": "23",
  "eventType": "ProductOfferingCreateEvent",
  "event": {
    "productOffering": {
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
        "@type": "OpenGatewayProductOffering",
        "@baseType": "DcsProductOffering",
        "lifecycleStatus": "launched",
        "version": "1.0.0",
```

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```
        "name": "Location Verification Antifraud CAMARA API Standalone",
        "description": "Location Verification product offering standalone, for
antifraud purpose, accessible with CAMARA API.",
        "lastUpdate": "2023-07-01T00:00:00Z",
        "allowedAction": [
          {
            "@type": "OpenGatewayAllowedProductAction",
            "id": "1",
            "action": "add",
            "@targetProductOrderItemSchema": {
              "@type": "ProductOrderItemAddLocationVerificationAntifraudStandalone",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ProductOrderItemAddLocationVerificationAntifraudStandalone"
            }
          },
          {
            "@type": "OpenGatewayAllowedProductAction",
            "id": "2",
            "action": "delete",
            "@targetProductOrderItemSchema": {
              "@type": "DcsProductOrderItemDelete",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
            }
          }
        ],
        "productSpecification": {
          "@type": "OpenGatewayProductSpecificationRef",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
          "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
          "name": "Location Verification Antifraud CAMARA API",
          "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
        },
        "productOfferingTermOrConditionSpecification": [{
          "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
          "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
          "name": "Location Verification Antifraud CAMARA API Standalone Product
Offering Terms and Conditions for ApplicationOwner",
          "description": "Terms and conditions to be approved by ApplicationOwner at
the order.",
          "attachment": {
            "@type": "OpenGatewayFileAttachment",
            "content": "JVBERi0x...DQolJUVPRg==",
            "mimeType": "application/pdf",
            "attachmentType": "termsAndConditions"
          }
        }],
        "productOfferingPrice": [
          {
            "id": "1",
            "@type": "OpenGatewayProductOfferingPrice",
            "lifecycleStatus": "active",
            "name": "Location Verification Antifraud CAMARA API Standalone Usage
Price",
            "description": "Location Verification for Antifraud standalone product
offering price. 8 EUR per API call for usage verify for FR, ES and GB. 8 USD per API
call for usage verify for US.",
            "lastUpdate": "2023-09-01T00:00:00Z",
```


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```
            "priceType": "composite"
          }
        ]
      }
  }
}
```

Delete Event

Notification for a deletion of a ProductOffering

```
Content-Type: application/json
{
  "@type": "OpenGatewayProductOfferingDeleteEvent",
  "correlationId": "13654384719",
  "domain": "Open Gateway Catalog",
  "timeOccurred": "2024-11-25T14:04:27Z",
  "eventId": "24",
  "eventType": "ProductOfferingDeleteEvent",
  "event": {
    "productOffering": {
        "@type": "OpenGatewayProductOffering",
        "@baseType": "DcsProductOffering",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb"
    }
  }
}
```

Attribute Value Change Event

Notification for an attribute value change of a ProductOffering

```
Content-Type: application/json
{
  "@type": "OpenGatewayProductOfferingAttributeValueChangeEvent",
  "@baseType": "DcsProductOffering",
  "correlationId": "13654384715",
  "domain": "Open Gateway Catalog",
  "timeOccurred": "2024-11-25T14:04:27Z",
  "eventId": "22",
  "eventType": "ProductOfferingAttributeValueChangeEvent",
  "event": {
    "productOffering": {
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
      "@type": "OpenGatewayProductOffering",
      "@baseType": "DcsProductOffering",
          "version": "1.1.0",
      "productOfferingPrice": [
        {
          "id": "1",
          "@type": "OpenGatewayProductOfferingPrice",
```

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```
          "description": "Location Verification for Antifraud standalone product
offering price, usage price. 9 EUR per API call for usage verify for FR, ES and GB.
9 USD per API call for usage verify for US."
        }
      ]
    }
  }
}
```


State Change Event

Notification for lifecycleStatus change of a ProductOffering


```
Content-Type: application/json
{
  "@type": "OpenGatewayProductOfferingStateChangeEvent",
  "correlationId": "13654384711",
  "domain": "Open Gateway Catalog",
  "timeOccurred": "2024-11-25T14:04:27Z",
  "eventId": "25",
  "eventType": "ProductOfferingStateChangeEvent",
  "event": {
    "productOffering": {
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
      "@type": "OpenGatewayProductOffering",
      "@baseType": "DcsProductOffering",
      "lifecycleStatus": "retired"
      }
    }
  }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on OpenGatewayProductOffering

Retrieves a productOffering by ID

`GET /productOffering/{id}?fields=…`

Description

This operation retrieves an OpenGatewayProductOffering entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a standalone productOffering

Request

```
GET /productOffering/5e9b10e5-d707-4a16-89d1-687c7f0bf81a
Content-Type: application/json
```

Response

```
200
{
```

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```
  "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/5e9b10e5-d707-4a16-89d1-
687c7f0bf81a",
  "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
  "lifecycleStatus": "launched",
  "version": "1.0.0",
  "name": "Open Gateway Location Verification bundle offer",
  "description": "Location Verification bundle offer. Includes Location Verification
CAMARA API for antifraud purpose and a bucket of 10000 API calls for verify location
usage, recurring monthly, usage non blocked when bucket is empty.",
  "lastUpdate": "2023-07-01T00:00:00Z",
  "allowedAction": [
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "1",
      "action": "add",
      "@targetProductOrderItemSchema": {
        "@type": "OpenGatewayProductOrderItemAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/OpenGatewayProductOrderItemAdd.schema.js
on#/definitions/OpenGatewayProductOrderItemAdd"
      }
  },
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "2",
      "action": "delete",
      "@targetProductOrderItemSchema": {
        "@type": "DcsProductOrderItemDelete",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/DcsProductOrderItemDelete.schema.json#/d
efinitions/DcsProductOrderItemDelete"
      }
    }
  ],
  "bundledProductOffering":[
    {
      "@type": "BundledProductOffering",
      "bundledProductOfferingOption":{
        "@type": "BundledProductOfferingOption",
        "numberRelOfferDefault":1,
        "numberRelOfferLowerLimit":1,
        "numberRelOfferUpperLimit":1
      },
      "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
      "@referredType": "OpenGatewayProductOffering",
      "version": "1.0.0"
    },
    {
      "@type": "BundledProductOffering",
      "bundledProductOfferingOption":{
        "@type": "BundledProductOfferingOption",
        "numberRelOfferDefault":1,
        "numberRelOfferLowerLimit":1,
        "numberRelOfferUpperLimit":1
      },
      "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
      "href": "https://mycsp.com/tmf-
```


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```
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
      "@referredType": "OpenGatewayProductOffering",
      "version": "1.0.0"
    }
  ],
  "productOfferingTermOrConditionSpecification": [{
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
    "name": "Open Gateway Location Verification bundle Product Offering Terms and
Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
      "@type": "OpenGatewayFileAttachment",
      "content": "cvUGFnZXMg...2l0ZUlkKD==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }],
  "productOfferingPrice": [
    {
      "id": "1",
      "@type": "OpenGatewayProductOfferingPrice",
      "lifecycleStatus": "active",
      "name": "Bundle price for Location Verification",
      "description": "One Time 3 EUR + recurring 10 EUR/month, including 100000 API
calls monthly + 9 EUR per Verify Location API call usage when usage bucket is over
until the bucket is refilled next month for FR, ES and GB. One Time 3 USD +
recurring 10 USD/month, including 100000 API calls monthly + 9 USD per Verify
Location API call usage for US.",
      "lastUpdate": "2023-09-01T00:00:00Z",
      "priceType": "composite"
    }
  ]
}
```

Retrieve a bundle productOffering

Request

```
GET /productOffering/2d4ef4d3-08ce-441d-ac76-133b6dad0ccb
Content-Type: application/json
```

Response

```
200
{
  "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
  "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
  "lifecycleStatus": "launched",
  "version": "1.0.0",
  "name": "Location Verification Antifraud CAMARA API Standalone",
```

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```
  "description": "Location Verification product offering standalone, for antifraud
purpose, accessible with CAMARA API.",
  "lastUpdate": "2023-07-01T00:00:00Z",
  "allowedAction": [
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "1",
      "action": "add",
      "@targetProductOrderItemSchema": {
        "@type": "ProductOrderItemAddLocationVerificationAntifraudStandalone",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ProductOrderItemAddLocationVerificationAntifraudStandalone"
      }
    },
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "2",
      "action": "delete",
      "@targetProductOrderItemSchema": {
        "@type": "DcsProductOrderItemDelete",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
      }
    }
  ],
  "productSpecification": {
    "@type": "OpenGatewayProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
  },
  "productOfferingTermOrConditionSpecification": [
    {
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
    "name": "Location Verification Antifraud CAMARA API Standalone Product Offering
Terms and Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
      "@type": "OpenGatewayFileAttachment",
      "content": "JVBERi0xL...DQolJUVPRg==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }
  ],
  "productOfferingPrice": [
    {
      "id": "1",
      "@type": "OpenGatewayProductOfferingPrice",
      "lifecycleStatus": "active",
      "name": "Location Verification Antifraud CAMARA API Standalone Usage Price",
      "description": "Location Verification for Antifraud standalone product
offering price. 8 EUR per API call for usage verify location for FR, ES and GB. 8
USD per API call for usage verify location for US.",
      "lastUpdate": "2023-09-01T00:00:00Z",
      "priceType": "composite"
```


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```
    }
  ]
}
```

Retrieve a bundled productOffering for API

Request

```
GET /productOffering/ab7792f8-6628-4c4b-a557-699adc26d4ce
Content-Type: application/json
```

Response

```
200
{
  "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
  "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
  "lifecycleStatus": "launched",
  "version": "1.0.0",
  "name": "Location Verification Antifraud CAMARA API Bundled",
  "description": "Location Verification product offering bundled, for antifraud
purpose, accessible with CAMARA API.",
  "lastUpdate": "2023-07-01T00:00:00Z",
  "allowedAction": [
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "1",
      "action": "add",
      "@targetProductOrderItemSchema": {
        "@type": "ProductOrderItemAddLocationVerificationAntifraudBundled",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/ProductOrderItemAddLocationVerificationAn
tifraudBundled.schema.json#/definitions/ProductOrderItemAddLocationVerificationAntif
raudBundled"
      }
    },
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "2",
      "action": "delete",
      "@targetProductOrderItemSchema": {
        "@type": "DcsProductOrderItemDelete",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
      }
    }
  ],
  "productSpecification": {
    "@type": "OpenGatewayProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
```

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```
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
  },
  "productOfferingTermOrConditionSpecification": [{
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
    "name": "Location Verification Antifraud CAMARA API Standalone Product Offering
Terms and Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
      "@type": "OpenGatewayFileAttachment",
      "content": "JVBERi0xL...DQolJUVPRg==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }]
}
```

Retrieve a bundled productOffering for usage volume

Request

```
GET /productOffering/0a2782b7-81a7-4af9-9e4f-d51bc690ec2d
Content-Type: application/json
```

Response

```
200
{
  "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
  "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
  "lifecycleStatus": "launched",
  "version": "1.0.0",
  "name": "Location Verification Verify Usage Volume Recurring Bundled",
  "description": "Location Verification Usage Volume of 10000 API calls for verify
location usage, included in a bucket, recurring monthly, usage non blocked when
bucket is empty.",
  "lastUpdate": "2023-07-01T00:00:00Z",
  "allowedAction": [
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "1",
      "action": "add",
      "@targetProductOrderItemSchema": {
        "@type": "OpenGatewayProductOrderItemAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/OpenGatewayProductOrderItemAdd.schema.js
on#/definitions/OpenGatewayProductOrderItemAdd"
      }
  },
```

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```
    {
      "@type": "OpenGatewayAllowedProductAction",
      "id": "2",
      "action": "delete",
      "@targetProductOrderItemSchema": {
        "@type": "DcsProductOrderItemDelete",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/DcsProductOrderItemDelete.schema.json#/d
efinitions/DcsProductOrderItemDelete"
      }
    }
  ],
  "productSpecification": {
    "@type": "OpenGatewayProductSpecificationRef",
    "id": "7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
    "name": "Location Verification Verify Usage Volume",
    "version": "1.0.0",
          "@referredType": "UsageVolumeProductSpecification"
  },
    "productOfferingTermOrConditionSpecification": [
    {
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
    "name": "Location Verification Verify Usage Volume Recurring Bundled Product
Offering Terms and Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
      "@type": "OpenGatewayFileAttachment",
      "content": "LjcNCiW...vUGFnZXMg==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }
  ]
}
```


List or find productOffering objects

`GET /productOffering?fields=…`

Description

This operation list OpenGatewayProductOffering entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

List all productOffering with lifecycleStatus launched

Request

```
GET /productOffering?lifecycleStatus=launched
Content-Type: application/json
```

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Response


```
200
[
  {
    "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
    "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
    "lifecycleStatus": "launched",
    "version": "1.0.0",
    "name": "Location Verification Verify Usage Volume Recurring Bundled",
    "description": "Location Verification Usage Volume of 10000 API calls for verify
location usage, included in a bucket, recurring monthly, usage non blocked when
bucket is empty.",
    "lastUpdate": "2023-07-01T00:00:00Z",
    "allowedAction": [
      {
        "@type": "OpenGatewayAllowedProductAction",
        "id": "1",
        "action": "add",
        "@targetProductOrderItemSchema": {
          "@type": "OpenGatewayProductOrderItemAdd",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/OpenGatewayProductOrderItemAdd.schema.js
on#/definitions/OpenGatewayProductOrderItemAdd"
        }
    },
      {
        "@type": "OpenGatewayAllowedProductAction",
        "id": "2",
        "action": "delete",
        "@targetProductOrderItemSchema": {
          "@type": "DcsProductOrderItemDelete",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/DcsProductOrderItemDelete.schema.json#/d
efinitions/DcsProductOrderItemDelete"
        }
      }
    ],
    "productSpecification": {
      "@type": "OpenGatewayProductSpecificationRef",
      "id": "7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
      "name": "Location Verification Verify Usage Volume",
        "version": "1.0.0",
          "@referredType": "UsageVolumeProductSpecification"
  },

    "productOfferingTermOrConditionSpecification": [
    {
    "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
    "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
    "name": "Location Verification Verify Usage Volume Recurring Bundled Product
Offering Terms and Conditions for ApplicationOwner",
    "description": "Terms and conditions to be approved by ApplicationOwner at the
order.",
    "attachment": {
```


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```
      "@type": "OpenGatewayFileAttachment",
      "content": "LjcNCiW...vUGFnZXMg==",
      "mimeType": "application/pdf",
      "attachmentType": "termsAndConditions"
    }
  }
  ]
    },
    {
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/2d4ef4d3-08ce-441d-ac76-
133b6dad0ccb",
      "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
      "lifecycleStatus": "launched",
      "version": "1.0.0",
      "name": "Location Verification Antifraud CAMARA API Standalone",
      "description": "Location Verification product offering standalone, for
antifraud purpose, accessible with CAMARA API.",
      "lastUpdate": "2023-07-01T00:00:00Z",
      "allowedAction": [
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "1",
          "action": "add",
          "@targetProductOrderItemSchema": {
            "@type": "ProductOrderItemAddLocationVerificationAntifraudStandalone",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ProductOrderItemAddLocationVerificationAntifraudStandalone"
          }
        },
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "2",
          "action": "delete",
          "@targetProductOrderItemSchema": {
            "@type": "DcsProductOrderItemDelete",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
          }
        }
      ],
      "productSpecification": {
        "@type": "OpenGatewayProductSpecificationRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "name": "Location Verification Antifraud CAMARA API",
        "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
      },
      "productOfferingTermOrConditionSpecification": [{
        "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
        "name": "Location Verification Antifraud CAMARA API Standalone Product
Offering Terms and Conditions for ApplicationOwner",
        "description": "Terms and conditions to be approved by ApplicationOwner at
the order.",
        "attachment": {
          "@type": "OpenGatewayFileAttachment",
```


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```
          "content": "JVBERi0xL...DQolJUVPRg==",
          "mimeType": "application/pdf",
          "attachmentType": "termsAndConditions"
        }
      }],
      "productOfferingPrice": [
        {
          "id": "1",
          "@type": "OpenGatewayProductOfferingPrice",
          "lifecycleStatus": "active",
          "name": "Location Verification Antifraud CAMARA API Standalone Usage
Price",
          "description": "Location Verification for Antifraud standalone product
offering price. 8 EUR per API call for usage verify for FR, ES and GB. 8 USD per API
call for usage verify for US.",
          "lastUpdate": "2023-09-01T00:00:00Z",
          "priceType": "composite"
        }
      ]
    },
    {
      "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/5e9b10e5-d707-4a16-89d1-
687c7f0bf81a",
      "@type": "OpenGatewayProductOffering",
  "@baseType": "DcsProductOffering",
      "lifecycleStatus": "launched",
      "version": "1.0.0",
      "name": "Open Gateway Location Verification bundle offer",
      "description": "Location Verification bundle offer. Includes Location
Verification CAMARA API for antifraud purpose and a bucket of 10000 API calls for
verify location usage, recurring monthly, usage non blocked when bucket is empty.",
      "lastUpdate": "2023-07-01T00:00:00Z",
      "allowedAction": [
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "1",
          "action": "add",
          "@targetProductOrderItemSchema": {
            "@type": "OpenGatewayProductOrderItemAdd",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/OpenGatewayProductOrderItemAdd.schema.js
on#/definitions/OpenGatewayProductOrderItemAdd"
          }
      },
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "2",
          "action": "delete",
          "@targetProductOrderItemSchema": {
            "@type": "DcsProductOrderItemDelete",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Customer/DCS/GSMAOperateAPI/DcsProductOrderItemDelete.schema.json#/d
efinitions/DcsProductOrderItemDelete"
          }
        }
      ],
      "bundledProductOffering":[
        {
          "@type": "BundledProductOffering",
          "bundledProductOfferingOption":{
            "@type": "BundledProductOfferingOption",
```


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```
            "numberRelOfferDefault":1,
            "numberRelOfferLowerLimit":1,
            "numberRelOfferUpperLimit":1
          },
          "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
          "@referredType": "OpenGatewayProductOffering",
          "version": "1.0.0"
        },
        {
          "@type": "BundledProductOffering",
          "bundledProductOfferingOption":{
            "@type": "BundledProductOfferingOption",
            "numberRelOfferDefault":1,
            "numberRelOfferLowerLimit":1,
            "numberRelOfferUpperLimit":1
          },
          "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
          "@referredType": "OpenGatewayProductOffering",
          "version": "1.0.0"
        }
      ],
      "productOfferingTermOrConditionSpecification": [{
        "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
        "id" : "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
        "name": "Open Gateway Location Verification bundle Product Offering Terms
and Conditions for ApplicationOwner",
        "description": "Terms and conditions to be approved by ApplicationOwner at
the order.",
        "attachment": {
          "@type": "OpenGatewayFileAttachment",
          "content": "cvUGFnZXMg...2l0ZUlkKD==",
          "mimeType": "application/pdf",
          "attachmentType": "termsAndConditions"
        }
      }],
      "productOfferingPrice": [
        {
          "id": "1",
          "@type": "OpenGatewayProductOfferingPrice",
          "lifecycleStatus": "active",
          "name": "Bundle price for Location Verification",
          "description": "One Time 3 EUR + recurring 10 EUR/month, including 100000
API calls monthly + 9 EUR per Verify Location API call usage when usage bucket is
over until the bucket is refilled next month for FR, ES and GB. One Time 3 USD +
recurring 10 USD/month, including 100000 API calls monthly + 9 USD per Verify
Location API call usage for US.",
          "lastUpdate": "2023-09-01T00:00:00Z",
          "priceType": "composite"
        }
      ]
    },
    {
      "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
      "@type": "OpenGatewayProductOffering",
```


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```
  "@baseType": "DcsProductOffering",
      "lifecycleStatus": "launched",
      "version": "1.0.0",
      "name": "Location Verification Antifraud CAMARA API Bundled",
      "description": "Location Verification product offering bundled, for antifraud
purpose, accessible with CAMARA API.",
      "lastUpdate": "2023-07-01T00:00:00Z",
      "allowedAction": [
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "1",
          "action": "add",
          "@targetProductOrderItemSchema": {
            "@type": "ProductOrderItemAddLocationVerificationAntifraudBundled",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/ProductOrderItemAddLocationVerificationAn
tifraudBundled.schema.json#/definitions/ProductOrderItemAddLocationVerificationAntif
raudBundled"
          }
        },
        {
          "@type": "OpenGatewayAllowedProductAction",
          "id": "2",
          "action": "delete",
          "@targetProductOrderItemSchema": {
            "@type": "DcsProductOrderItemDelete",
            "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/DcsProductOrderItemDelete.schema.json#/definitions/DcsPr
oductOrderItemDelete"
          }
        }
      ],
      "productSpecification": {
        "@type": "OpenGatewayProductSpecificationRef",
        "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "name": "Location Verification Antifraud CAMARA API",
        "version": "1.0.0",
          "@referredType": "ApiProductSpecification"
      },
      "productOfferingTermOrConditionSpecification": [{
        "@type": "OpenGatewayProductOfferingTermOrConditionSpecification",
        "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
        "name": "Location Verification Antifraud CAMARA API Standalone Product
Offering Terms and Conditions for ApplicationOwner",
        "description": "Terms and conditions to be approved by ApplicationOwner at
the order.",
        "attachment": {
          "@type": "OpenGatewayFileAttachment",
          "content": "JVBERi0xL...DQolJUVPRg==",
          "mimeType": "application/pdf",
          "attachmentType": "termsAndConditions"
        }
      }]
    }
]
```


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Operations on OpenGatewayProductSpecification

Retrieves a productSpecification by ID

`GET /productSpecification/{id}?fields=…`

Description

This operation retrieves an OpenGatewayProductSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a productSpecification by id

Request

```
GET /productSpecification/4b6591ef-5ede-4885-9543-0c5e9070ade9
Content-Type: application/json
```

Response

```
200
  {
        "@type": "ApiProductSpecification",
        "@baseType": "OpenGatewayProductSpecification",
        "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "name": "Location Verification Antifraud CAMARA API",
        "version": "1.0.0",
        "lifecycleStatus": "active",
        "description": "Location Verification product specification, for antifraud
purpose, accessible with CAMARA API.",
        "lastUpdate": "2023-09-23T16:42:23Z",
        "targetProductSchema": {
          "@type": "ApiProductLocationVerificationAntifraud",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraud.schema.json
#/definitions/ApiProductLocationVerificationAntifraud"
        },
        "apiStandardName": "CAMARA",
        "apiVersionInformation": [
          {
            "@type": "ApiVersionInformation",
            "apiName": "location-verification",
            "apiVersion": "1.0.0",
            "apiBasePath": "/location-verification/v1",
            "apiStatus": "active",
            "apiGrantInformation": [
              {
                "@type": "ApiGrantInformation",
                "purpose": "dpv:FraudPreventionAndDetection",
                "legalBasis": "dpv:Consent",
                "scope": [
```

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```
                  "location-verification:verify"
                ],
                "grantType": [
                  "authorization_code",
                  "urn:openid:params:grant-type:ciba"
                ]
              }
            ]
          }
        ],
        "attachment": [
          {
            "@type": "OpenGatewayURLAttachment",
            "url":
"https://mycsp.com/OpenGateway/documentation/LocationVerificationDeveloperGuide.html
",
            "attachmentType": "developerDocumentation",
            "mimeType": "text/html"
          }
        ]
      }
```

List or find productSpecification objects

`GET /productSpecification?fields=…`

Description

This operation list OpenGatewayProductSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

List all productSpecification with lifecycleStatus launched

Request

```
GET /productSpecification?lifecycleStatus=active
Content-Type: application/json
```

Response

```
200
[
  {
      "@type": "UsageVolumeProductSpecification",
      "@baseType": "OpenGatewayProductSpecification",
      "id": "7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/7407a0fd-9a01-47a2-b32a-1ab85bde3d32",
      "name": "Location Verification Verify Usage Volume",
        "version": "1.0.0",
        "lifecycleStatus": "active",
        "description": "Location Verification Usage Volume of 10000 API calls for
```

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```
verify location usage, included in a bucket, recurring monthly, usage non blocked
when bucket is empty.",
        "lastUpdate": "2023-09-23T16:42:23Z",
        "targetProductSchema": {
          "@type": "UsageVolumeProductLocationVerification",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/UsageVolumeLocationVerification.schema.js
on#/definitions/UsageVolumeProductLocationVerification"
            },
        "productSpecificationRelationship":
        [
          {
            "@type": "OpenGatewayProductSpecificationRelationship",
            "id":"4b6591ef-5ede-4885-9543-0c5e9070ade9",
            "relationshipType": "requires",
            "@referredType":"ApiProductSpecification"
          }
        ]
  },
  {
        "@type": "ApiProductSpecification",
        "@baseType": "OpenGatewayProductSpecification",
        "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/4b6591ef-5ede-4885-9543-0c5e9070ade9",
        "name": "Location Verification Antifraud CAMARA API",
        "version": "1.0.0",
        "lifecycleStatus": "active",
        "description": "Location Verification product specification, for antifraud
purpose, accessible with CAMARA API.",
        "lastUpdate": "2023-09-23T16:42:23Z",
        "targetProductSchema": {
          "@type": "ApiProductLocationVerificationAntifraud",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ApiProductLocationVerificationAntifraud"
        },
        "apiStandardName": "CAMARA",
        "apiVersionInformation": [
          {
            "@type": "ApiVersionInformation",
            "apiName": "location-verification",
            "apiVersion": "1.0.0",
            "apiBasePath": "/location-verification/v1",
            "apiStatus": "active",
            "apiGrantInformation": [
              {
                "@type": "ApiGrantInformation",
                "purpose": "dpv:FraudPreventionAndDetection",
                "legalBasis": "dpv:Consent",
                "scope": [
                  "location-verification:verify"
                ],
                "grantType": [
                  "authorization_code",
                  "urn:openid:params:grant-type:ciba"
                ]
              }
            ]
          }
        ],
        "attachment": [
          {
```


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```
            "@type": "OpenGatewayURLAttachment",
            "url":
"https://mycsp.com/OpenGateway/documentation/LocationVerificationDeveloperGuide.html
",
            "attachmentType": "developerDocumentation",
            "mimeType": "text/html"
          }
        ]
      }
]
```

Operations on ApiProductSpecification

See operations on OpenGatewayProductSpecification.

Operations on UsageVolumeProductSpecification

See operations on OpenGatewayProductSpecification.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "@type": "Hub",
    "callback": "https://in.listener.com",
    "query":
"eventType=ApiProductOfferingCreate,ApiProductOfferingStateChange,ApiProductOffering
VersionCreate"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "@type": "Hub"
```

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```
    "id":"42",
    "href": "tmf-api/openGatewayOperateAPIProductCatalog/v5/hub/42"
    "callback": "https://in.listener.com",
    "query":
"eventType=ApiProductOfferingCreate,ApiProductOfferingStateChange,ApiProductOffering
VersionCreate"
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

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Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	June 2025	Olivier Arnaud (Orange) olivier.arnaud@orange.com	
Version 5.0.0-beta1 (preview)	March 2025	Olivier Arnaud (Orange) olivier.arnaud@orange.com	

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